Basics of sustainability

Foundations and challenges of sustainability

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Preface

This introduction to the foundations and challenges of sustainable development is aimed at students and lecturers who deal with the most pressing issues of our time from different disciplinary perspectives. It arose from the desire to promote a common understanding - across disciplinary boundaries, in teaching, research and social practice.

The world as we know it today is characterised by a multitude of interwoven crises: Climate change, loss of biodiversity, social inequalities, economic instability and political polarisation - in short, a polycrisis. These challenges are neither linear nor easy to solve. They call for a profound rethink of our economic, social and ecological systems, our actions and our idea of development and progress.

Against this background, this textbook sees itself as a shared learning space: it offers systematic access to central concepts, scientific perspectives, normative questions and specific fields of action for sustainable development. The aim is not only to impart knowledge, but also to promote the ability to reflect, think critically and be creative.

This book was written in the context of the study programmes of the Centre for Development and Environment (CDE) at the University of Bern. It reflects the many years of experience in research and teaching on sustainable development and brings together contributions from various disciplines - from environmental sciences, geography and economics to sociology, ethics and law.

Special thanks go to all the authors who have contributed to the creation of this book with their expertise, passion and commitment. We would also like to thank the students, whose critical questions and perspectives are a key driver for the further development of the content. After all, sustainable development is not only a scientific project, but above all a social project - and this begins with education.

We hope that this book will provide a sound basis for your own engagement with sustainability - and encourage you to take responsibility and play an active role in shaping a fair and sustainable world.

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Introduction

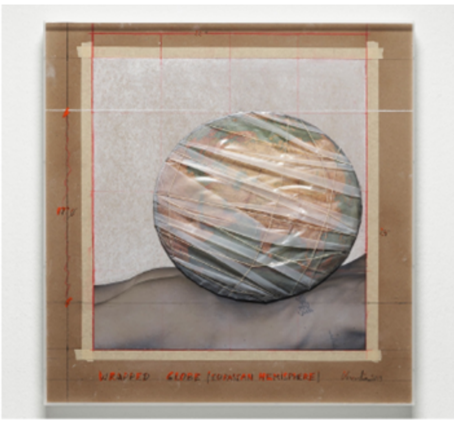

Figure 1: “Wrapped Globe (Eurasian Hemisphere)” by Christo (2019)

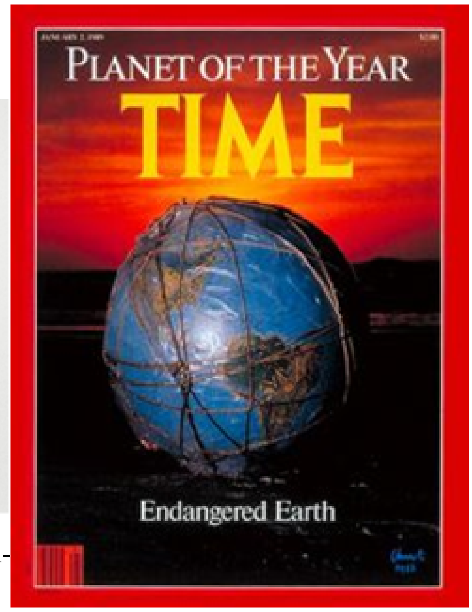

Figure 2: “Planet of the Year” Times Magazine Cover (1989)

Amid the mounting challenges of sustainable development, Christo and Jeanne-Claude’s “Wrapped Globe” is a powerful symbol of humanity’s responsibility towards our planet and its resources. The artwork depicts a globe wrapped in transparent plastic and a filigree net. Meanwhile, in real life, the world is facing a “polycrisis” – the word used to describe the many serious crises our Earth is facing, including ecological crises, growing inequality, excessive national debt, and the effects of the Covid-19 pandemic, to name but a few. In a polycrisis, crises are increasingly intertwined and mutually reinforcing (Tooze 2022), and they are mainly caused by a structural dependence on growth (as measured by gross domestic product, GDP) (Hickel et al. 2022); a vicious circle of ever-increasing concentration of economic and political power in the hands of a few (Piketty 2014); and persistent inequalities between and within countries (Chancel et al. 2021; Milanović 2016). And yet, we keep striving for GDP growth in our society and our economy, in order to maintain and create jobs, finance our social security systems, secure tax revenues, and fulfil the needs of companies and industries that depend on growth to exist. As these expectations become increasingly unrealistic, the idea of decoupling economic growth from resource consumption has gained traction. However, there is no empirical evidence that doing so will achieve anywhere near the scale required to halt multidimensional ecological collapse (Hickel and Kallis 2020; Parrique et al. 2019; Wiedmann et al. 2020).

Full and empty world

The plastic cover and the net wrapped around Christo's globe thus represent the interconnectedness and interdependence of the Earth's various elements, and emphasize the need to maintain and preserve these relationships. A similar idea was described by the economist Herman H. Daly (2015), who put forth a concept of the "empty" and the "full" world. The empty world describes a situation in which human activities and resource use have only a minor impact on the environment. In this world, natural systems are still intact and untouched, and resources are sufficient to meet human needs. This contrasts with the full world, in which human activities and resource use overload the ecosystem and pollute the environment. Since at least the Second World War, we have pursued an industrialized society and a growth economy. As a result, we now live in a "full" world, where natural resources are scarce and the balance of ecosystems is under threat.

Daly's epiphany came in 1962 upon reading Rachel Carson's book, *Silent Spring*, which called for a life in harmony with nature. Daly was already sceptical about the hyper-individualism of economic models, and Carson's work highlighted the conflict between a growing economy and a fragile environment. Following a lecture by the economist Nicholas Georgescu-Roegen on his magnum opus, *The Entropy Law and the Economic Process* (1971), Daly adopted the idea that the economy was more like an hourglass than a pendulum, with valuable resources turning into waste and thus largely irreversibly lost.

There is no master plan to counter the polycrisis and make our economic and social system more sustainable. To contribute to sustainable development, we need to analyse the causes of symptoms such as biodiversity loss or global warming, and learn to understand how they are connected and how they interact. In our study programmes, we will therefore look at current problems and challenges such as global warming, pollution, biodiversity loss, social inequality, and economic disparities, to understand how they can be tackled at both the local and the global level.

The aim of this textbook is to encourage readers to think critically about the role of individuals, communities, businesses, and governments in the context of sustainable development – and thus to identify and pursue approaches to support sustainable development. In addition, we want to develop visions of the future, especially in the Master's study programmes, that make it possible to imagine a high quality of life in a sustainable modern age, and that make changing the present seem attractive rather than daunting. We want to envision a different food culture, a different economic system, a different type of land use, and a different way of building and living. To make progress towards sustainability, it will be critical to engage stakeholders and foster collaboration across sectors and disciplines. As this textbook will emphasize, education, communication, and participation will also be indispensable in shaping a more sustainable future.

With Christo's Wrapped Globe in mind, and Daly's understanding of the empty and the full world as a foundation, we will embark on a journey through the many aspects of sustainable development. We hope that you will find this journey both informative and inspiring, and that it will give you an understanding of both the urgency and the opportunities for sustainable development.

Finally, we hope that this textbook will inspire readers to reflect on their own roles and responsibilities in relation to sustainability, and that it will equip you to make a difference to ensure that the Earth remains a place worth living in for future generations. Only together can we bring about the changes needed to create a sustainable, just, and environmentally friendly world.

Part I

Foundations

Chapter 1

Understandings and concepts of sustainability

Where do the concept, mission statement, and guiding principles of sustainable development come from? How did the concept emerge, how did it evolve – and why? The following explanations provide an overview of the history of sustainability, the political background, and key conferences and documents. In short, what has made sustainability what it is today. It's about seeing the big picture. Because only those who know the past can assess the future.

Sustainability has been the buzzword of recent years, from science to politics and business to the media. Initially, the term has a positive connotation because it is associated with the long term, the durable, and many things are sustainable today: coffee, corporate philosophy, even tuna pizzas. These are the demands of a consumer society with an inconsiderate appetite for abundance. First thing in the morning, the shampoo removes our dandruff "sustainably". Then we drive to work at a company that prides itself on its "sustainable corporate philosophy", and over lunch we discuss sustainable investments. At home, we stick a tuna pizza in the oven – sustainably sourced and produced, of course, as it says on the box.

Today, the term "sustainability" is everywhere: it's used in connection with energy, mobility, building renovation, nutrition, population development, corporate environmental management, and climate protection. It's also used in art, culture, design, and advertising.

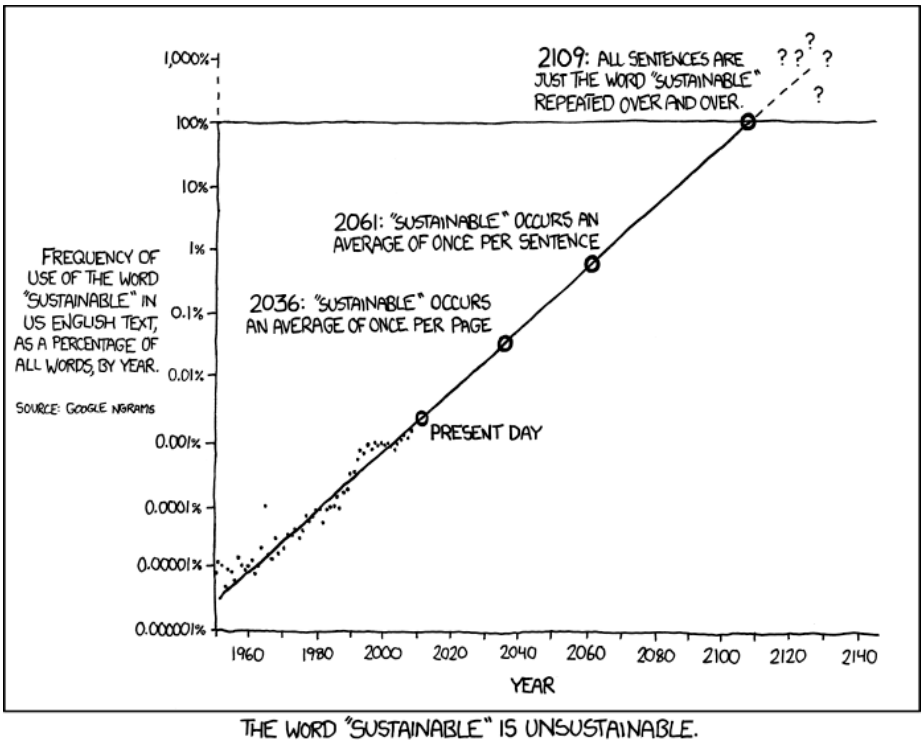

Figure 1.1: Frequency of use of the word "sustainable" in US English texts, as a percentage of all words per year, based on data from Google Ngrams. Source: https://xkcd.com/1007/

Is the term "sustainability" so overused as to slowly become meaningless? Even if this is the case, we shouldn't abandon key terms like this lightly. For example, it's still important for companies to have a sustainability strategy, even if many such strategies are inadequate or amount to greenwashing. We need to clarify the meaning of "sustainability" and hold those who use it to account. And we should base our interpretation on science and historical developments. The historical precursors to the sustainability model explained in this textbook are:

- Start of the discussion about sustainability: Carlowitz's forest management principle of 1713
- Clash of economy and ecology First and second UN Development Decades (1960s and 1970s)
- The Limits to Growth (1972)
- Brundtland Report (1987)
- Rio Earth Summit 1992 Agenda 21 UN Millennium Development Goals (MDGs) (2000–2015)
- 2030 Agenda with the Sustainable Development Goals (SDGs) (2015–2030)

1.1 A normative concept

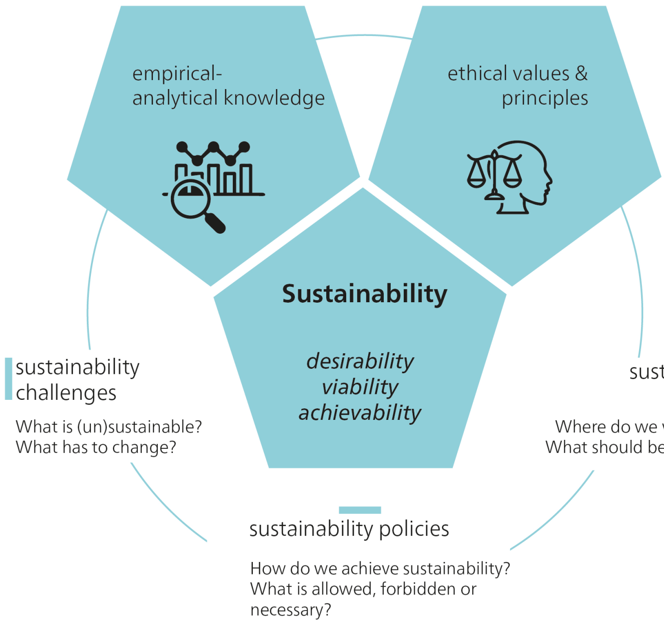

Figure 1.2: Sustainable Development as a normative concept. Source: Own illustration.

To achieve sustainability, we need * *sustainable development**: a strategic process that requires changes in our socio-ecological systems and institutions. "Development" implies controlled improvement, and the ethical question of what exactly "better" means is key. The use of the term "development" has been criticized by some (e.g. Lang and Mokrani 2013; Kothari et al. 2019), as it is often equated with unbridled growth. Unbridled growth of the "ecological footprint" – the proportion of the biosphere used for human production, consumption, and waste – is unsustainable in the long term. Nonetheless, there are different interpretations of "development", ranging from economic growth to improving quality of life. Sustainable development therefore remains a stimulating and controversial concept. Sustainability and sustainable development play a key role in today's political discourse. The term "unsustainability" refers to conditions or developments that are considered negative, while "sustainability" represents a positive state.

The concept of sustainable development commits us to certain values and norms that define ethical goals and rules of behaviour. These values and norms shape our ideas of what we consider a desirable or "positive" change. A neutral point of view is not possible, as our perceptions are shaped by a variety of influences. Our personal experiences, upbringing, social environment, and cultural background all influence how we see and interpret the world, including our perceptions of what "should be" and what actions "should be taken". Ethics play a key role in determining how the current situation can be improved to achieve sustainability, by defining normative goals and limits. Sustainable development is therefore a conceptual framework that is strongly driven by ethical considerations.

Sustainability challenges such as poverty reduction and climate change are therefore ethical challenges. The identification of situations as sustainability problems (and therefore as "negative") and the choice of solutions are based on ethical values. Sustainability goals are also based on ethical values, as well as on knowledge of cause-and-effect relationships. Sustainability issues are often referred to as "wicked problems", because of their complex and pluralistic nature. Global warming, in particular, has been described as a "super wicked problem", as finding solutions is urgent, political institutions are often inadequate, and decision-making processes suffer from short-sightedness.

1.2 Global challenges as wicked problems

Mike Hulme, author of *Why We Disagree About Climate Change* (2009), suggests that we view climate change as a "wicked problem" of enormous scale and likely longevity. This approach helps us see climate change not as a problem that needs to be solved, but as a condition in which we are directly involved. The categorization of problems as "wicked", i.e. those that do not lend themselves to clear-cut solutions, originated with the planning theorists Horst Rittl and Melvin Webber (1973). They argued that planners have to deal with unpredictable human behaviour, and that some problems are too complicated to be solved completely. Some "wicked problems" may never be fully solved, but we can learn to cope with them and not let them dominate us.

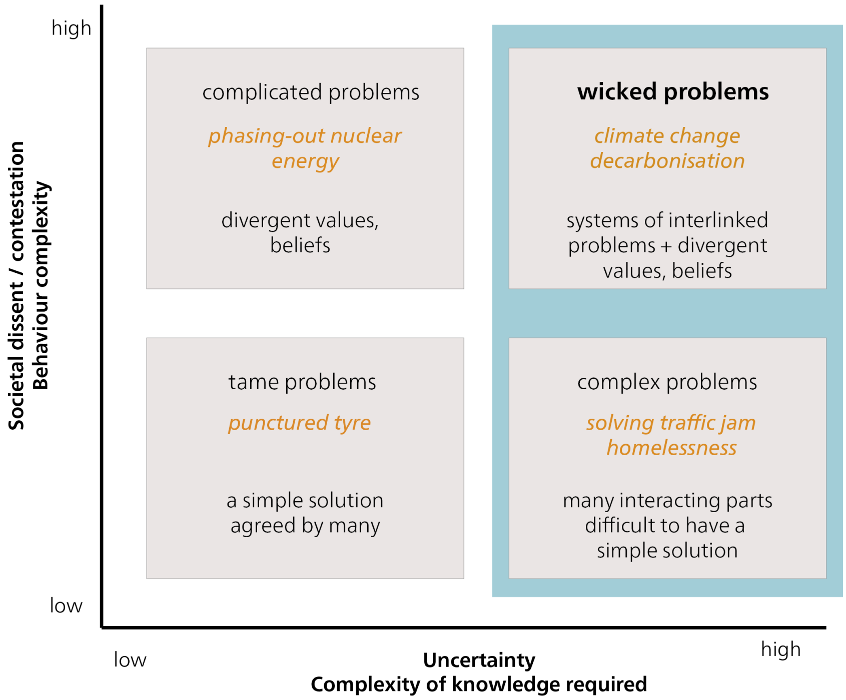

Figure 1.3: Sustainability challenges as wicked problems. Source: Own illustration based on Roth, G. L., & Senge, P. M. (1996).

Wicked problems

According to Bannink and Trommel (2019), wicked problems arise at the intersection of factual uncertainty and a heterogeneity of preferences and interests. Wicked problems are characterized by (Alford and Head 2017; Rittel and Webber 1973; Sediri et al. 2020):

- **Complexity**: Wicked problems are characterized by many interrelated factors and interactions. There are no clear cause-and-effect relationships, and changes in one area can have unforeseen effects in other areas.
- **Normative conflicts**: Wicked problems are perceived and interpreted differently by different stakeholders and interest groups. There is no clear definition or consensus on what exactly the problem is, or how it should be solved.
- **Interdisciplinarity**: Wicked problems require an interdisciplinary approach as they involve different topics, perspectives, and stakeholders. Finding a solution often requires the cooperation and coordination of different disciplines and experts. Uncertainty: Incorrect, missing, or inaccessible information about the problem situation and about the continuity of the values of the variables involved.

- **No definitive solution**: Wicked problems defy clear-cut solutions. They are dynamic, change over time, and require continuous adjustment and iterations.

Examples of wicked problems include climate change, poverty alleviation, global health, and sustainability. The complexity and interaction of different factors in these areas make it difficult to find simple and clear-cut solutions. Dealing with wicked problems requires a high degree of reflection, collaboration, and the use of systemic thinking methods.

How have we manoeuvred ourselves into the current situation, where wicked problems pose such a challenge to sustainable development? In this textbook, we will analyse and learn to understand three key global challenges:

- the emergence of human-induced global warming;
- the persistence of global poverty and rising inequalities;
- and the threat of species extinction and overexploitation of natural resources.

The debate about human influence on global warming and climate change has intensified since the publications of the Intergovernmental Panel on Climate Change (IPCC). The driving force behind global warming is our dependence on oil and other fossil fuels for transport, (electricity) generation, agriculture, and many everyday products. Another wicked problem is global poverty. Although there have been global efforts to reduce poverty for decades, the globalized market economy often leads to increased poverty in certain regions, and the gap between rich and poor seems to be widening overall (Chancel et al. 2021). High-income countries have maintained their prosperity and living standards at the expense of others. These are "externalized costs", and they include environmental degradation. For example, the wealth and living standards of high-income countries are largely responsible for the threat of species extinction and the overuse of natural resources (Lessenich 2018). In addition to the above mentioned "big three" – climate change, poverty, and biodiversity loss – there are many other global sustainability challenges, such as deforestation, desertification, declining soil fertility, dwindling fish stocks, pollution, wars, conflicts, and increasing migration.

In analysing these comprehensive challenges we must also consider the temporal dimension. In 2004, graphs depicting socio-economic and Earth system trends from 1750 to 2000 were first published, revealing a dramatic upsurge and aptly termed "The Great Acceleration" (Steffen et al. 2004).

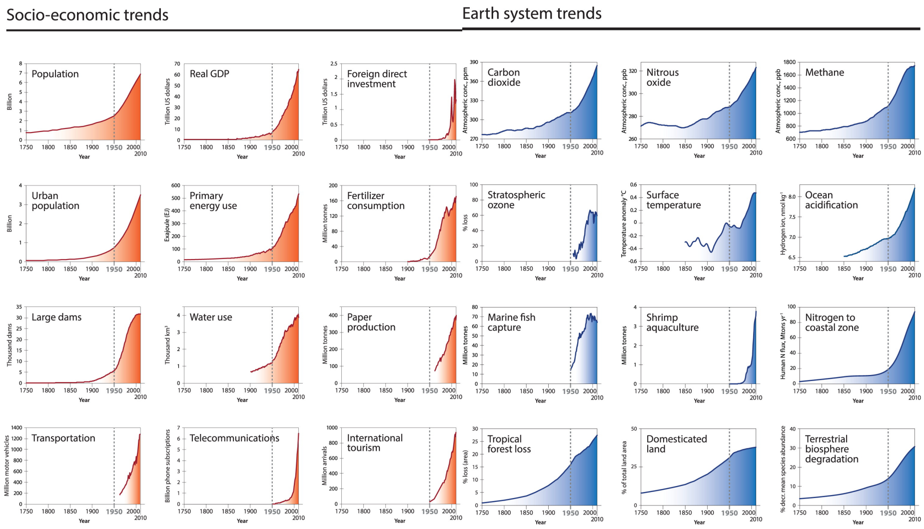

Figure 1.4: 12 socio-economic and 12 Earth system trends from 1750 to today are a strong indication that the Earth system has entered a new state. Source: https://futureearth.org/2015/01/16/the-great-acceleration/.

The Great Acceleration described the exponential growth dynamics that occurred as a result of the Industrial Revolution, which led to a surge in productivity in the mode of production and a significant increase in material wealth. At the same time, but to a much lesser extent, there was a massive increase in natural resource consumption and emissions. Socio-economic growth went hand in hand with the acceleration of biophysical trends. Figure **??** describes some examples of important biophysical and socio-economic indicators, all of which start to increase with the Industrial Revolution. From the middle of the 20th century, the trend towards exponential growth becomes apparent.

> The Wheat and Chessboard Problem: an illustration of exponential growth
>
> The Wheat and Chessboard Problem is a mathematical problem often used to illustrate the concept of exponential growth. In the story, a servant asks the king to fill every square on a chessboard with grains of wheat, doubling the number on each square. Growth starts slowly, but with each doubling, the number of grains increases exponentially. By the 50th square, the number of rice grains would be large enough to cover Berlin's 365-metre-high television tower on Alexanderplatz. This story illustrates the immense power of exponential growth and its impressive results. Understanding the concept of exponential growth is crucial to analysing phenomena such as population growth, technological progress, environmental change, and the spread of disease. It illustrates how even small changes or developments in a system can have a significant impact. History reminds us that we need to think carefully about how we manage such growth and the long-term consequences it can have.

Resource extraction has increased significantly in recent decades, from 22 billion tonnes in 1970 to 70 billion tonnes in 2010 (UNEP 2016). Global warming is another pressing issue. The latest IPCC report (2023) predicts an increase in global average temperature of between 1.5 and 5.8 degrees Celsius, depending on the scenario and future emissions. Another alarming phenomenon is deforestation. Every two seconds, an area of forest the size of a football field is cut down – an area the size of New York City every day. And then there is the decline in biodiversity, marked by the extinction of animal and plant species, which threatens the ecological diversity of our planet.

The Great Acceleration (Figure **??**) describes the observable and measurable (negative) developments of key socio-economic and biophysical indicators since 1950. The causes of these negative developments can be far removed from the place where the problems arise or become most evident. For example, job losses in one country may be caused by a multinational company's decision to relocate part of its operations to another country, in order to maximize profits. We cannot hope to fix such issues unless we understand how the system – in this case, the globalized economy – works. Similarly, we need to understand the Earth's global climate systems to imagine what the consequences of global warming might be in a particular area or region.

This is why many of the methods and concepts of sustainability science require systemic thinking. A systemic thinking approach often begins with brainstorming to create a "rich picture" of all the factors you need to consider in order to understand the current behaviour of the system in question. For example, in the case of a multinational company closing a particular business unit, these would be the factors that might influence the decision to continue, expand, or close certain business units. While an initial mapping might look overly complex or even messy, creating a linear flowchart can help clarify the flows of inputs and influences. This kind of mapping may already reveal opportunities to reassess the relevant influences and redesign the system to avoid unwanted outcomes. However, further work may be required to develop "conceptual models" of the system that identify unforeseen opportunities for improvement.

Incorporating systemic thinking into the methods and concepts of sustainability science enables us to tackle the complexity of problems and develop sound strategies for sustainable development. Systemic thinking thus provides an important basis for understanding and shaping the different understandings and concepts of sustainable development, as the next chapter discusses in more detail.

1.3 Sustainability: When did we start talking about it?

In 1713, Saxony's chief mining officer, Hans Carl von Carlowitz, demonstrated the necessity and possibility of "sustainable forestry" in a book entitled *Sylvicultura oeconomica, oder hauswirthliche Nachricht und Naturmässige Anweisung zur wilden Baum-Zucht (Sylvicultura oeconomica, A Guide to the Cultivation of Native Wild Trees)*. Another sustainability pioneer was Duchess Anna Amalia, the mother of Duke Carl August. In the late 1700s, she initiated the world's first forestry reform, which was aimed at ensuring a continuous and steady supply of timber. At that time, Europe had an insatiable appetite for materia prima, or raw materials needed for building ships and houses, and for cooking and heating. This demand threatened to deplete resources and endanger their long-term survival.

In the 18th century, scholars in Germany as well as in other parts of Europe addressed the finite nature of natural resources. Unlike von Carlowitz, however, no one spoke about sustainability. One important aspect was the provision of food for a growing population. Before industrialization, economic growth was largely determined by nature as a factor of production, such as the number of mines or ships built, or the amount of land available for farming. However, Thomas Robert Malthus, a British economist, warned that food production would not be able to keep up with the rapid population growth following the Industrial Revolution. Malthus believed that the population would grow at a exponential rate, while food production would increase at a linear rate. Although Malthus's thesis did not come true in the dramatic way he predicted, his work is often regarded as the first systematic treatise on the limits to growth in a finite world.

For a time, Malthus's ideas fell out of favour, as unlimited growth seemed possible due to the Industrial Revolution and the rise of capitalism. Talk of natural limits didn't resurface until the mid-20th century, when a "neo-Malthusian" perspective re-emerged in the context of the Club of Rome (founded in 1968), debates on planetary boundaries, and critiques of growth. While Malthusianism traditionally focuses on external, physical limits to growth, the ecological economist, Giorgos Kallis, proposes a different approach. Kallis (2019) advocates personal moderation and voluntary restraint, an approach to life that has deep roots in Romanticism and ancient Greek philosophy. He argues for an inner limit to our wants, to free us from the economic assumption of scarcity and the associated obsession with growth. While acknowledging the reality of external limits, Kallis believes they should not dominate the environmental argument against limitless growth. Kallis therefore asks whether it makes sense to frame climate change primarily as a problem of external limits, of scarcity, or of a finite atmosphere that cannot absorb any more of our emissions.

In the Malthusian logic, we would have to ever increase our production to meet the needs of a world with limits. As long as this mindset persists, the focus is on how we can exceed those limits, and how we can push the capacity of the climate to absorb a growing population and its demands.

1.3.1 The clash of economy and ecology

Originally, sustainability was a principle of resource economics that combined the economic goal of maximizing long-term forest use with ecological conditions for regrowth. In the 19th century, the concept of an 'eternal forest' emerged with the aim of putting an end to overexploitation and ensuring a long-term supply of wood, a vital raw material. Forestry scientists such as Heinrich Cotta developed mathematical methods for calculating timber stocks and growth. Over time, however, the focus shifted towards the pure yield theory, which focused on short rotation periods, high-yield monocultures and maximizing financial benefits. The focus was suddenly on the highest possible direct cash yield instead of a steady high timber yield. Natural cycles were replaced by capitalist dynamics; exchange value took precedence over utility value. This devalued the guiding principle of sustainability, and it took over a hundred years, until the 1960s and 70s, for the scientific disciplines of ecology and sustainability to regain prominence.

1.3.2 Club of Rome and The Limits to Growth

> "If the present growth trends in world population, industrialization, pollution, food production, and resource depletion continue unchanged, the

limits to growth on this planet will be reached sometime within the next one hundred years." (Meadows et al. 1972, 23)

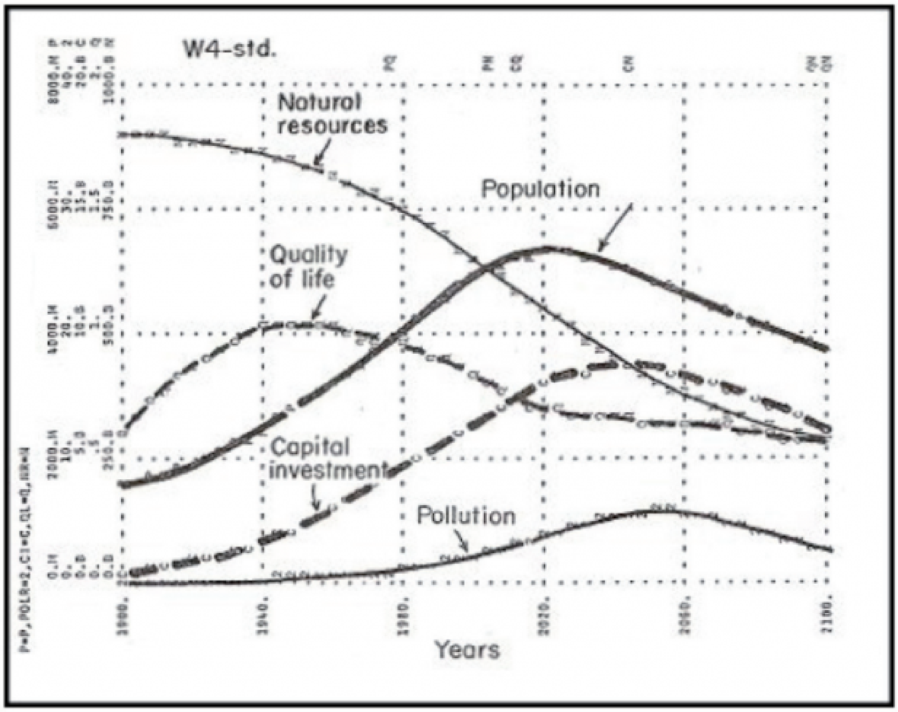

Figure 1.5: The "Basic model" of the simulation carried out at MIT, published in the autumn of 1970. Source: https://donellameadows.org/.

In 1972, the Club of Rome published a report, *The Limits to Growth*, which introduced a much broader understanding of "sustainability". Scientists begin to call for a global state of equilibrium, or homeostasis, that would only be possible through coordinated international action. They integrate economic, ecological, and social aspects of sustainability and use the model of the dynamics of complex systems ("system dynamics") to understand the world. They take into account the interactions between population density, food resources, energy, materials, capital, environmental degradation, and land use. Computer simulations of different scenarios show similar results: a catastrophic decline in the world's population and living standards within 50 to 100 years, if current trends continue. The problem with the resource- and emission-intensive industrialized societies is that growth is not linear but exponential. In the long run, this kind of growth leads to collapse. And ecological collapse can only be prevented through a course correction, argued the report.

The *Meadows Report*, as it is also known, was heavily criticized for its predictions and methods. Some critics argued that the system dynamics model was too simplistic and failed to consider important factors such as technology and innovation. Others said insufficient account was taken of human adaptability and the possibility of political solutions and change. Others yet criticized what they called a Malthusian outlook and a pessimistic view of the future. Despite these criticisms, however, the report sparked an important debate about sustainability and the need to protect the environment.

A year later, E.F. Schumacher published *Small is Beautiful: Economics as if People Mattered* (1973). Schumacher challenged the prevailing notion of limitless economic growth and technological progress. Instead, he advocated for a sustainable, human-centred economy that respects local communities and the environment. Schumacher argued that a decentralized economy, based on human values, not just profit, would create a better future for all. He also emphasized the importance of education and cultural development in creating a sustainable society.

1.4 From the development debate to the sustainability debate

Worsening air and water pollution helped raise the profile of environmental issues in politics and the media. Greenpeace was founded in 1971. In the 1960s and 1970s, Paul Crutzen and his research team studied the impact of nitrogen oxides on the ozone layer, predicting that this layer would be greatly depleted by human-made CFCs. The use of CFCs in refrigerators and air conditioners was subsequently banned. In response to the growing importance of environmental issues, the United Nations organized the first ever major environmental conference in Stockholm in 1972. The United Nations Conference on the Human Environment, as it was called, led to the creation of the United Nations Environment Programme (UNEP) and independent environment ministries in many countries.

Figure 1.6: Demonstration against air pollution, Zurich, December 1986. Source: Schweizerisches Sozialarchiv/Gertrud Vogler.

The emergence of environmental problems in the 1970s and 1980s coincided with the first “development crises” after the Second World War. In those days we still spoke of "underdeveloped countries". Following the success of the Marshall Plan in rebuilding Europe after World War II, it was assumed that similar programmes, such as a Marshall Plan for Africa, would lead to a rapid catch-up in socio-economic development of the poor countries of the Global South. But the 70s and 80s proved otherwise. Even exemplary countries such as Mexico and Brazil succumbed to debt crises, demonstrating that achieving socio-economic development and overcoming poverty were far more complex challenges.

From then on, it was clear that issues related to development and the environment had to be considered together at the intergovernmental level. In 1983, the United Nations set up a World Commission on Environment and Development, chaired by the Norwegian Prime Minister, Gro Harlem Brundtland.

1.4.1 Brundtland Commission

Gro Harlem Brundtland appointed 22 commission members, ¾ of whom were from the Global South. During this time, the Soviet Union, facing economic stagnation, was beginning to change political course. As the arms race with the US had contributed to a substantial budget deficit, Mikhail Gorbachev, the newly appointed General Secretary of the Communist Party of the Soviet Union, introduced major reforms. At the same time, a new world view and global consciousness were starting to emerge. It was against this background that the Brundtland Commission was tasked with drawing up a report on the perspective of global, sustainable, environmentally friendly development up to and beyond the year 2000. The report presented in 1987 was entitled *Our Common Future*, but it's often referred to as the *Brundtland Report*.

The Brundtland Report found that global environmental problems were mainly due to unsustainable consumption and production patterns in the North and severe poverty in the South. The overexploitation of nature and depletion of natural resources were exacerbating inequalities (of income and wealth), increasing absolute poverty, and posing a threat to peace and security. The Report sought to develop a fair and just definition of sustainability:

> "Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs." (WCED 1987)

This definition introduced an ethical perspective into the sustainability debate, placing the principle of responsibility at the centre for both present and future generations. By focusing on human needs, the Brundtland Commission adopted an anthropocentric position. The Brundtland Commission identified three key principles for analysing problems and guiding action: a global perspective, the interconnectedness of environment and development, and the pursuit of justice, or equity. The concept of justice/equity was further divided into two distinct perspectives:

1. The intergenerational perspective: Responsibility for future generations.
2. The intragenerational perspective: Responsibility for people living today, particularly in developing countries, and ensuring equity within countries.

> Sustainability and sustainable development
>
> The concepts of "sustainability" and "sustainable development" differ in focus. Sustainability is static; it emphasizes a consistent state, while sustainable development is the dynamic process of achieving that state, implying movement and referring to something that is emerging.

1.5 From Rio 1992 to today

After the Brundtland Report's call for international action, the focus turned to translating demands and proposals into binding treaties and conventions. The UN chose a conference as an instrument for this, and it was held exactly 20 years after 1972 Stockholm Conference, the first global environmental conference. The UN Conference on Environment and Development held in Rio de Janeiro, Brazil, in June 1992 – also known as the Earth Summit – was the largest international conference to date, with delegates from over 170 nations.

The Rio Earth Summit adopted the following five documents:

- A Forest Declaration, aimed at the ecological management and protection of the world's forests;
- A Climate Protection Convention committing states to reducing greenhouse gas emissions worldwide to 1990 levels;
- A Biodiversity Convention to combat the decline in biodiversity through steps that are binding under international law;
- The Rio Declaration on Environment and Development; and Agenda 21, the best known of the five agreements, according to which national governments are responsible for implementing the sustainability model in their countries.

The Rio Declaration on Environment and Development emphasizes that long-term economic progress is only possible using an ecosystem approach. This, in turn, requires a new and equitable global partnership among governments, people, and key societal groups. To achieve this, international agreements are necessary to protect the environment and the development system. The Rio Declaration established key principles of sustainable development, including precaution and polluter pays. For example, Principle 15 states: "In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation" (UN 1992).

In December 1992, the Commission on Sustainable Development (CSD) was established to ensure effective follow-up of the Summit.

1.5.1 Agenda 21: Think globally – act locally

Agenda 21, which contains 40 chapters, addresses all critical policy areas and actions for sustainable development. According to the Preamble, "…integration of environment and development concerns and greater attention to them will lead to the fulfilment of basic needs, improved living standards for all, better protected and managed ecosystems and a safer, more prosperous future. No nation can achieve this on its own; but together we can – in a global partnership for sustainable development." (UN CSD 1992)

1.5.2 The Millenium Development Goals (MDGs)

The core principles of Agenda 21 – empowering women, protecting the environment, and promoting an equitable and inclusive society – laid the foundation for the Millennium Development Goals (MDGs) . Adopted by the UN in 2000, the MDGs comprised eight specific goals to be achieved by 2015. Although not all of these goals were met, the MDGs raised significant awareness of the need for sustainable development. The MDGs were succeeded by the Sustainable Development Goals (SDGs), which were adopted by the UN in 2015.

1.5.3 Rio+20

The UN Conference on Sustainable Development, or Rio+20, was held in Rio de Janeiro in 2012, 20 years after the historic 1992 Earth Summit that adopted Agenda 21. The main objective of Rio+20 was to reaffirm global political commitment to sustainable development. Participants, including government leaders and NGO and private sector representatives, discussed a wide range of issues, including poverty eradication , environmental protection, sustainable energy, food security, and resource management. The Summit's key outcome was the adoption of a declaration entitled "The Future We Want". This declaration renewed the international community's commitment to sustainable development and to the Rio Principles and past action plans , and outlined further measures to achieve these goals. In addition, Rio +20 paved the way for a universal development framework to define global sustainability goals and priorities beyond 2015.

1.5.4 The 2030 Agenda and the Sustainable Development Goals (SDGs)

> "We are the first generation that can put an end to poverty and we are the last generation that can put an end to climate change, so we [must] address climate change." Ban-Ki Moon, UN Secretary General 2007–2016, on the 2030 Agenda

Rio +20 laid the foundation for the 2030 Agenda, which the UN adopted in 2015. The 2030 Agenda builds on the results of Rio+20 and sets out a comprehensive framework for sustainable development. The 2030 Agenda is based on five key principles or pillars (the 5 Ps) and introduces the 17 Sustainable Development Goals (SDGs), which aim to create a sustainable and just world by 2030.

The 5 Ps of the SDGs

The 5 Ps represent five key principles of sustainability : People, Planet, Prosperity, Peace, and Partnership.
People: This P stands for the aim to promote social justice, equal opportunities, good health, and education for all people. It also seeks to end poverty, promote gender equality, and improve people's well-being.
Planet: The aim of this P is to protect natural resources and to use them sustainably, to preserve biodiversity, to protect the climate, and to reduce pollution. Overall, it seeks to preserve our planet and promote sustainable environmental practices.
Prosperity: This refers to economic growth, sustainable production and consumption patterns, and the creation of jobs and economic prosperity for all. This P aims to promote a strong and sustainable economy that benefits all people.
Peace: This is about promoting peace, justice, good governance, and strong institutions. It's also about preventing conflict, reducing violence, and building inclusive societies.
Partnership: This refers to the importance of global cooperation, partnerships, and solidarity among all stakeholders. It's about working together to implement the 2030 Agenda, and sharing resources, experience, and knowledge.

1.5.5 2015 Paris Climate Agreement

The Paris Climate Agreement of 2015 is a landmark international agreement adopted at the 21st Conference of the Parties (COP 21) to the UN Framework Convention on Climate Change (UNFCCC) in Paris. The Agreement aims to limit the global temperature rise to well below 2 degrees Celsius compared to pre-industrial levels, and to endeavour to limit it to 1.5 degrees Celsius. The Paris Agreement is based on the principle of common but differentiated responsibilities and respect for national circumstances. It encourages all countries to take action to reduce their greenhouse gas emissions and to set themselves voluntary climate targets, known as Nationally Determined Contributions (NDCs).

The 2015 Paris Climate Agreement applies to every country that has ratified it, including Switzerland . As a Party to the Agreement, Switzerland has committed to contributing to the global reduction of greenhouse gas emissions and to implementing the Agreement's objectives.

1.5.6 Conclusion – there is progress, but it's too slow

While there has been some progress in institutionalizing and disseminating sustainability approaches in business, policymaking, and other areas, a large number of studies show that the world as a whole is on an unsustainable course (see the Voluntary National Review of Switzerland 2022). A similar picture is painted by various studies on the global environmental situation, such as UNEP's Global Environment Outlook 5 (GEO-5) published in 2012, the 2010 Millennium Development Goals Report (MDGs Report 2010), and reports on the 2030 Agenda and climate change (IPCC 2022). As these studies show, the international community is a long way from sustainable, intergenerationally equitable ecological, social, and economic development.

For example, international climate policies have not stopped the rise in CO_2 emissions, the main cause of human-induced climate change, which have risen by around 50% since 1990. Species loss continues to accelerate in many places. The global fight against poverty falls short of the desired goals, and economic globalization over the past three decades has worsened socio-economic inequality in many countries, often linked to sociopolitical and sociocultural disparities.

1.6 The guiding principle of sustainable development as a normative concept

Turning the concept of sustainability into action and developing implementation strategies is a major challenge for our society. There are different understandings and concepts of sustainability and sustainable development, with no consensus on how to achieve sustainability and which measures to prioritize. Some of these understandings are presented below. The diversity of approaches reflects the different perspectives and interests of different stakeholders. To navigate this complexity, we need to reconcile economic, social, and environmental considerations. This requires an integrative approach. The challenge is to define common goals and develop strategies to achieve sustainability at all levels – global, national, regional, and local. We need broad societal participation and dialogue and cooperation between governments, business, civil society, and research institutions, to find solutions and drive the necessary change.

> "When powerful metaphors become fashionable buzzwords, we risk that diversity is accompanied by vagueness, i.e. the phenomenon of a term that has several meanings that 'have so much in common that it is difficult to separate them' " (Strunz 2012, 113; as cited in Feola 2015, 377)

The concept of sustainability usually has a positive connotation, but it can be viewed from different perspectives. In order to function as a guiding principle (in an ecological, social, and economic sense), it requires clear criteria. But what grounds can we use to define these criteria?

According to Hirsch Hadorn and Brun (2007), "sustainable development" should:

- Fulfil *needs*…
- …in a *just* way,
- …with a view to people living today and in the future, and
- taking into account the diversity of values and the limits to which nature can be used.

The guiding principle of sustainable development is therefore not only the result of scientific research, but is first and foremost a normative, ethically based concept. It brings together "ethical and analytical ideas" and formulates "norms that express what is desirable and what should happen" (Renn et al. 2007, 39). As a result, sustainable development is a social process of negotiation and decision-making – of searching, learning, and gaining experience – that is guided by ethical considerations. Accordingly, sustainability research must always be aware of its involvement in social processes of perception and evaluation.

1.6.1 What values and norms should we be guided by, and why?

Moving from the concept of sustainable development to its implementation, ethics come into play. Ethics provide evaluation criteria, methodological procedures, or principles for "justifying and criticizing rules of action or normative statements about how we should act" (Fenner 2008, 5). Ethics also help to structure and justify complex decisions that arise in dilemma situations. In contrast to problems with single solutions, a dilemma is more complex and involves trade-offs and the weighing up of different options for action. Ethics provide orientation and decision-making structures that provide a framework for finding a suitable course of action in such situations. The core function of ethics is therefore not to solve monocausal problems, but to structure and categorize complex dilemmas.

Put simply, "morality" refers to personal or social beliefs about right and wrong, while "ethics" is the philosophical study of these beliefs. Ethics studies the rules and principles that "ought" to govern human behaviour. Ethics can be divided into general and applied ethics (Figure **??**).

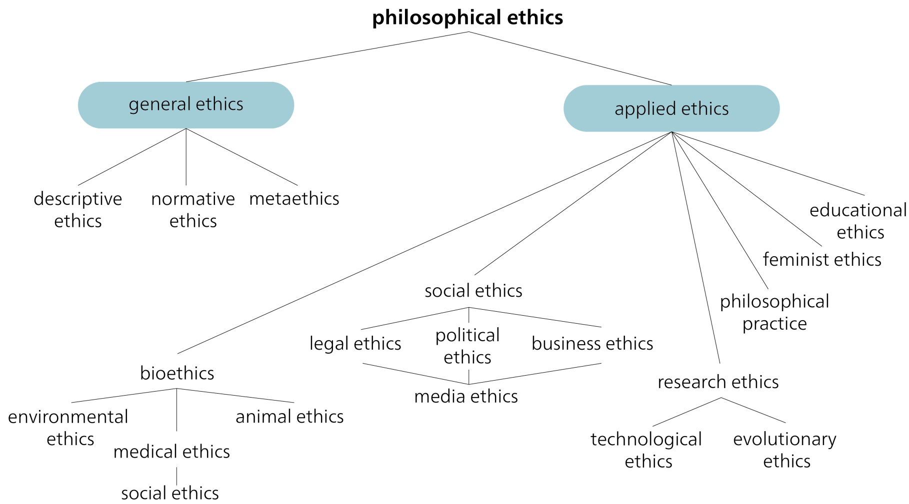

Figure 1.7: The different branches of philosophical ethics. Source: Own illustration based on Pieper and Thurnherr 1998, 9.

General ethics provides us with methods and concepts to discuss fundamental moral problems. It consists of three branches: normative ethics, descriptive ethics, and metaethics. Normative ethics focuses on determining what is morally right or wrong.

For example, questions about what constitutes a good life can have different answers, depending on people's individual values.

For this reason, normative ethics is further divided into teleological and deontological approaches. Teleological ethics (rooted in *telos*, the Greek word for "end", "purpose", or "goal") evaluates actions based on the good they produce. Utilitarianism, for example, is a teleological theory that evaluates actions based on the resulting utility or happiness. One of the first systematic elaborations of utilitarianism is Jeremy Bentham's (1748–1832) An *Introduction to the Principles of Morals and Legislation* (1780). Deontological ethics, on the other hand, evaluates actions on the basis of their characteristics rather than their consequences. Its root is deon, Greek for "duty". An example of a deontological ethical theory is Kantian ethics, developed by the German philosopher Immanuel Kant.

The distinction between teleological and deontological approaches is often discussed using the "trolley problem" (see "Justice: What's The Right Thing To Do? Episode 01 'THE MORAL SIDE OF MURDER.'" 2009; Sandel 2013 for further reading).

Further reading

Justice: What's The Right Thing To Do? Episode 01 "THE MORAL SIDE OF MURDER." 2009. With Michael Sandel. The Moral Side of Murder. https://www.youtube.com/watch?v=kBdfcR-8hEY.
Sandel, Michael. 2013. *Gerechtigkeit.* Ullstein.

1.6.2 The imperative of responsibility – Hans Jonas

Hans Jonas's "imperative of responsibility" (1979) offers an important contribution to the sustainability debate by emphasizing our ethical responsibility for future generations and for nature. Jonas's approach, which has been described as a kind of "ethics of the future", aims to address the new challenges posed specifically by a technological civilization. The link to sustainability ethics lies in his understanding of the concept of ethical responsibility towards future generations, which he views as an asymmetric, non-reciprocal relationship (Jonas 1987, 177). The asymmetry is derived from the power of a "moral subject" over someone or something that requires care; Jonas describes the vulnerability of life (ontological vulnerability) and responsibility as our duty of care to protect other beings from harm (Jonas 1987, 391).

Jonas's approach differs from other ethical theories in that it doesn't take the existence of humanity for granted, and instead includes duties towards future generations as well as to nature. Jonas argues that the first duty of ethics of the future consists in recognizing the distant effects of human action (Jonas 1987, 64). Given the uncertainty about the long-term consequences of human action, he proposes a decision-making principle based on the Latin term, in dubio pro malo, which he interprets to mean: "[…] when in doubt, give the worse prognosis precedence over the better, as the stakes have become too high for the game" (Jonas 1987, 67). Jonas thus develops a new categorical imperative, emphasizing the need to preserve both nature and humanity. It is based on the idea that nature has inherent value and purpose: "Act in such a way that the effects of your action are contractual with the permanence of genuine human life on earth", or "Act in such a way that the effects of your action are not destructive to the future possibilities of such life" (Jonas 1987, 36).

Already in the late 1970s, therefore, Hans Jonas articulated a fundamental uncertainty about the risks and harm that could arise for future generations from the technologies that were being used, such as nuclear energy. Decision-making around issues such as new technologies requires a continuous evaluation of risks and opportunities, often amid incomplete information and uncertain future forecasts. These choices not only impact the environment, but also raise ethical questions about our obligation to ensure sustainability for future generations. Jonas's imperative of responsibility emphasizes the link between responsibility and duty as key concepts in the sustainability debate. According to Jonas, current generations have a forward-looking responsibility to future generations. He leaves open, however, the extent of this duty, and whether future generations are entitled to absolute or comparative standards.

Jonas's imperative of responsibility can be considered in connection with various areas of responsibility that represent different ethical approaches: anthropocentrism, pathocentrism, biocentrism, and physiocentrism.

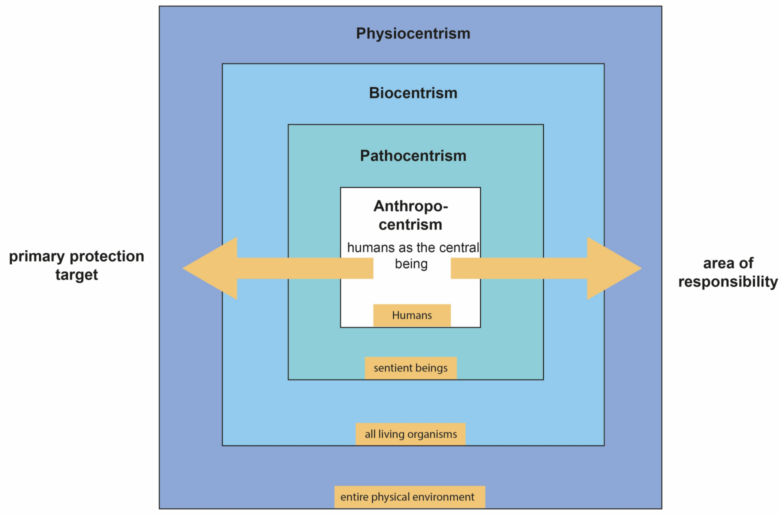

Figure 1.8: Different ethical approaches. Source: Own illustration based on Carnau (2011, 143).

Anthropocentrism views humans as the central beings, emphasizing responsibility towards human society and its well-being. This includes considering the needs and interests of both present and future generations. Pathocentrism, on the other hand, emphasizes responsibility towards sentient beings and their well-being. It recognizes that not only humans, but also animals and other living beings, have the right to be free from suffering and harm. Biocentrism, in turn, recognizes the intrinsic value of all living organisms and focuses on the protection and preservation of biodiversity. It emphasizes the importance of respecting the natural environment and promoting sustainable practices. Finally, physiocentrism focuses on the responsibility towards the entire physical environment, including inanimate nature and ecosystems. It recognizes the intrinsic value of nature and emphasizes the complex interactions between all components of the ecosystem.

Hans Jonas's imperative of responsibility reconciles all of these different areas of responsibility. It urges us to take responsibility for people as well as for other living beings, nature, and the environment. In doing so, it is important to adopt a balanced and comprehensive perspective that takes into account long-term effects and the well-being of all.

By integrating these different areas of responsibility, we can cultivate a holistic and sustainable ethic that takes due account of the needs and interests of all aspects of life and nature.

1.7 Sustainable Development in the context of development debates

Is "sustainable development" a development theory? Yes and no. No, because sustainable development is a broader social model and acts as an overarching framework that encompasses various development theories. Yes, because sustainable development is a normative theory that de facto competes in scientific and social practice with other theories of social and, in particular, socio-economic development.

Development theories analyse social development, either retrospectively or prospectively, and either as a whole or through specific aspects (e.g. economic development). The retrospective study of social development seeks to explain past or current social developments and to draw lessons for future planning and development policy. The prospective study of social development aims to inform social development management and policy. Development theories can be categorized into different types: normative, strategic, and explanatory theories. Each type focuses on different aspects and objectives of development research.

Normative theories focus on the question of what development should look like, and what normative goals it should achieve. They examine values and goals associated with development, often referring to concepts such as social justice, sustainability, or poverty alleviation. Normative theories emphasize the definition of criteria for good development, and provide a normative framework for policy decisions and measures. Examples include the theory of sustainable development, John Rawls's theory of social justice, or Amartya Sens's Capability approach.

Strategic theories investigate which strategies and measures are needed to achieve desired development outcomes. They focus on questions of policy design, resource allocation, and the implementation of measures, analysing how specific approaches, such as neoclassical growth theory, can promote development.

Explanatory theories aim to explain the causes and mechanisms of development. They analyse factors that drive development processes in societies and study the links between the different variables. This involves examining economic, social, political, or cultural factors to identify patterns and connections. Examples include modernization theory and dependency theory.

Note: All theories are based on normative foundations – i.e. underlying values and beliefs – and therefore imply, even unintentionally, certain ideas about how society should develop.

Table 1.1: Overview of Development Debates. Source: Extended table based on Egli et al. (2022), p. 378

	1950s/60s	1970s	1980s	1990s	Since 2008
Narrative	Economic growth, modernization, and integration into the world economy	Fight against extreme poverty and inequality	Lost decade: collapse of commodity prices; increase in debt	Sustainable development; decade of hope	Polycrisis
Basic idea	Catch-up development through industrialization and large-scale projects	Satisfaction of basic needs, growth, and effective distribution	Overcoming the debt crisis through structural adjustment measures, reduction of state services	Global and sustainable environmental policy, economic growth, peacekeeping, continuing basic needs strategy, participation	Bringing together various crisis phenomena (financial crisis, climate change, biodiversity crisis, etc.) and examining their complex interrelations; search for integrated solutions
Theories	Modernization theories (stage theories)	Dependency theories versus modernization theories	Market liberalism and neoliberalism	- Theories of sustainable development - Neoclassical growth theory - Postcolonial theories - Capability approach - Post-development theories	Additional resilience theories, post-growth theories

	1950s/60s	1970s	1980s	1990s	Since 2008
Goals	Geopolitical classification of the "Third World"; containment of communism, modernization of agriculture, rapid industrialization and technological progress, opening up of new markets	Aid for self-help, appropriate technologies, rural development, promotion of women	Increase in exports, balanced state budgets	Paradigm shift in development aid towards development cooperation, environmental and social compatibility of development, ensuring access to resources and infrastructure for all	
Consequences	Economic dependence and indebtedness increase, income disparities widen, poverty rises, ecological damage due to overuse and inappropriate technologies, undemocratic regimes are supported for geopolitical reasons	Selective progress in education and health, problems of debt and environmental damage	Living standards of the poorest population groups deteriorate severely, environmental exploitation, debt spiral, increase of dependency of the Global South on Global North countries	Focus shifts from purely economic development to human development; industrialized countries are obliged to implement sustainable development; emergence of a global development partnership	

Development theories

Modernization theory (1950s–1980s) emphasizes the transition from traditional societies to modern, industrialized societies as the basis for development. For proponents of modernization theory, the countries of the Global South are developing in the same direction as industrialized countries, but at a much slower pace. Modernization theory values economic growth, technological progress, and institutional adjustments. It is often associated with stage theories, which emerged in the 1950s and 1960s. Stage theories view development as a gradual process characterized by clearly defined stages or phases of development, with emphasis on economic growth and modernization. Prominent proponents include Walt Rostow, who identified five stages of development in his work, The Stages of

Economic Growth: A Non-Communist Manifesto (i.e. traditional society, preconditions for take-off, take-off, drive to maturity, and mass consumption).
Dependency theory (1970s and 1980s) emphasizes the structural dependence and inequality between developed and developing countries. It argues that developing countries are trapped in an unjust global economic system and remain dependent on developed countries.
Neoclassical growth theory (as of the 1990s) focuses on the connection between capital accumulation, technological progress, and economic growth. It emphasizes free markets, trade, and investment as drivers of development.
Postcolonial theories (as of the 1990s) emphasize the historical and structural inequality between former colonial powers and colonies. They argue that development can only be achieved through a critical examination of the legacy of colonialism.
Sen's Capability approach (as of the 1990s) emphasizes the importance of freedom, social justice, and human development. It emphasizes the necessity to improve the opportunities and capabilities of people in order to achieve development.
Post-development theories (as of the 1990s) criticize linear development models that impose Western ideas of progress, emphasizing instead local practices and values. Prominent thinkers in this field include Arturo Escobar, who critically reflects on the idea of development in his work Encountering Development: The Making and Unmaking of the Third World, and Vandana Shiva, whose Decolonizing the North calls for a sustainable and fairer development policy in the Global South that reduces the influence of the Global North.

1.8 Sustainability models

1.8.1 Three dimensions of sustainability

The established three-sector model of sustainability, introduced in the 1987 Brundtland Report, s eeks to balance environmental, economic, and social concerns. This model, which later also became known as the "triple bottom line" model, emphasized that economic development should be profitable while ensuring that environmental and social impacts remain neutral. On closer inspection, however, the model – initially depicted as a building with three pillars – has a major weakness. While it was intended to suggest that if any one pillar was weak, the system as a whole would be unsustainable - it could be argued that removing one of the pillars – or even the two outer pillars – would not make the building collapse if the remaining pillar(s) were strong enough. Despite its widespread use and overall success, the model struggles with transparency and equity. For example, measuring economic success requires quantitative indicators, while it's hard to find similar metrics for social well-being, especially emotional aspects. Some critics argue that the model prioritizes conventional economic thinking, neglecting important social aspects. An overemphasis on human economic systems can also prevent us from seeing our dependence on non-human ecological systems.

1.8.2 Sustainability as an intersection

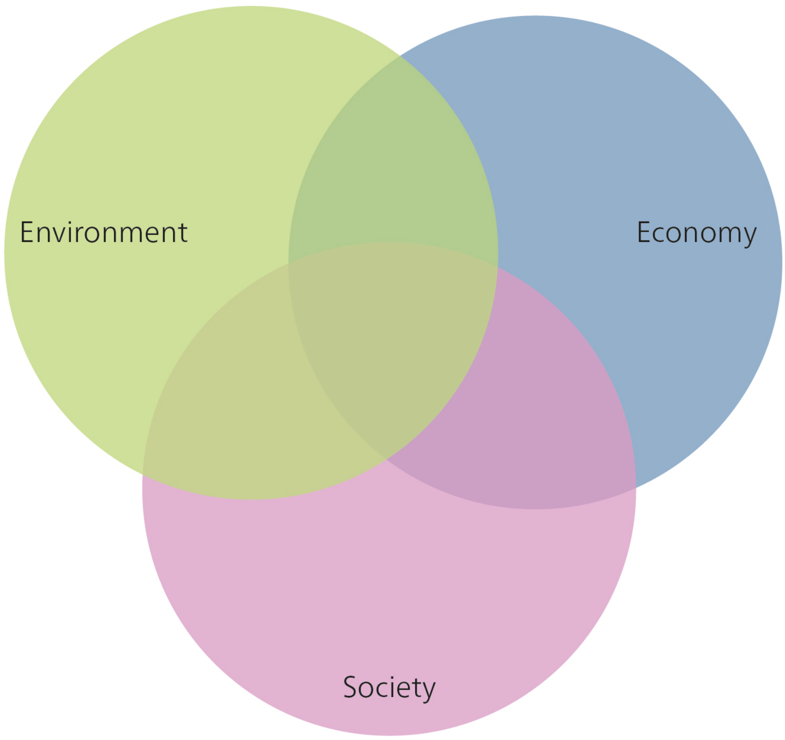

Figure 1.9: Sustainability as an intersection. Source: Own illustration.

A model displaying the three sectors as a Venn diagram (overlapping rings or circles, sometimes known as the intersection or triad model) offers an alternative to the separate pillars, emphasizing the interconnectedness of the three dimensions. This model illustrates that there can be a closer connection between two areas, and that the boundaries are fluid.

1.8.3 Nested sustainable development

Giddings, Hopwood, and O'Brien (2002) challenge the idea of separate spheres for the economy, society, and environment. They propose a "nested" model, where the economy is seen as part of society, which in turn is embedded in the environment. This model, they argue, ensures that economic decisions are always constrained by social and environmental factors. Global ecosystems form the basis for social systems (i.e. society). Embedded therein is the economy, which serves society's needs. A potential drawback of this model, however, is that if the economy is the starting point for all sustainability considerations, there's a risk of prioritizing it over social and environmental considerations.¨

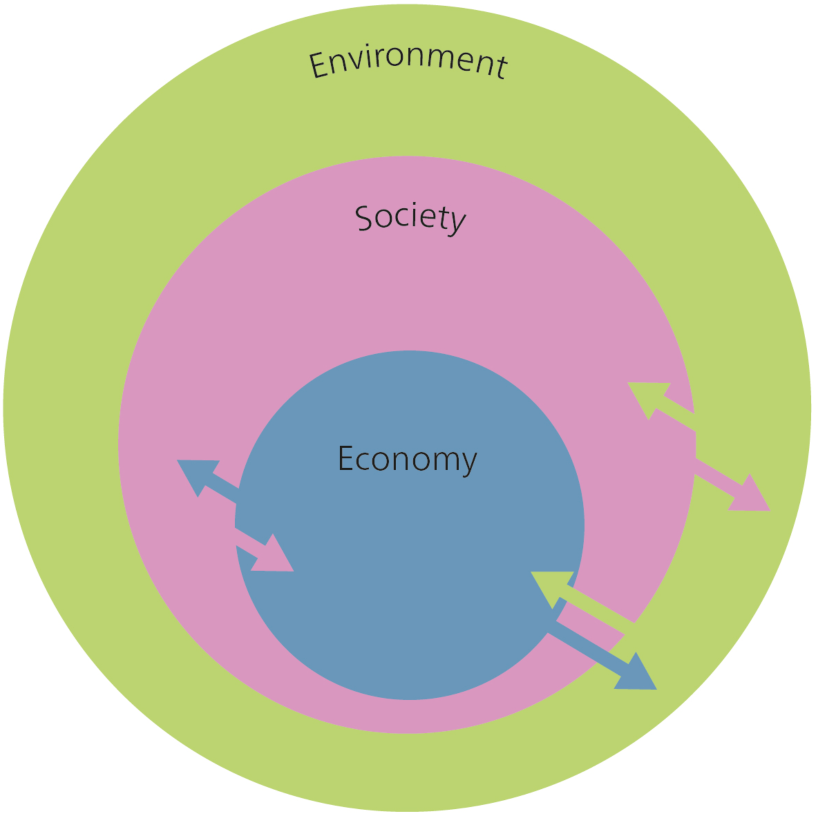

Figure 1.10: The nested model of sustainability. Source: Own illustration based on Giddings, Hopwood, and O'Brien (2002, 192)).

1.8.4 A four-dimensional model of sustainability

One problem with the three-sector model of sustainability is that that it's unclear what falls within the broad "social" sphere. It has therefore been suggested that a fourth dimension be added, to emphasize certain aspects that might otherwise remain hidden in the social sector. For example, Jon Hawkes, an Australian cultural commentator, has suggested adding "cultural vitality" as a fourth pillar of sustainability, in addition to environmental, economic, and social concerns.

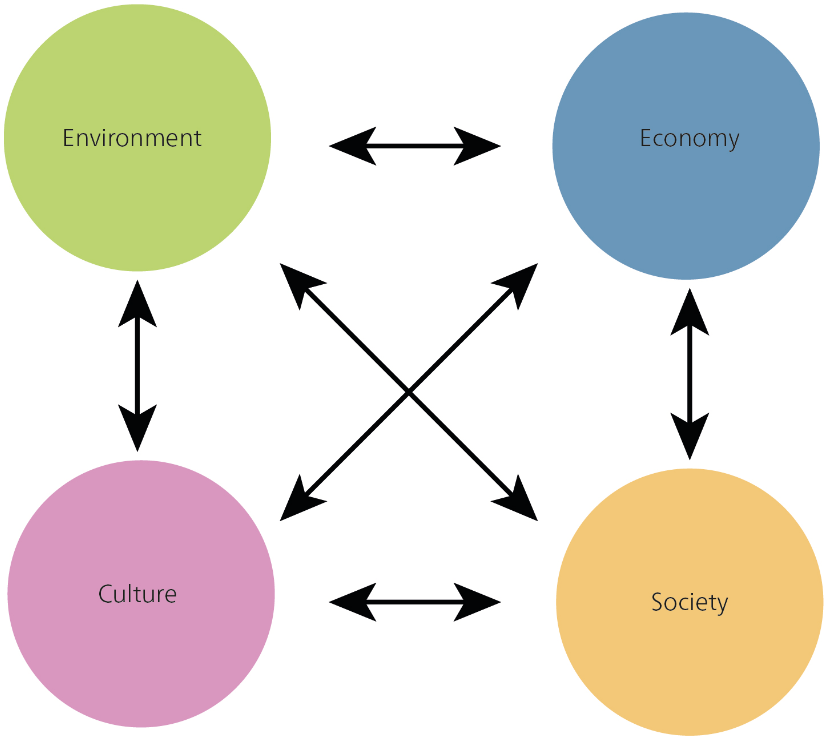

Figure 1.11: Four-dimensional model of sustainability. Source: Own illustration, adapted from Hawkes, 2004.

Cultural sustainability is often seen as a part of social sustainability. Sometimes the term "sociocultural sustainability" is also used. In this context, "culture" refers to aspects such as equity, opportunities for participation, awareness of sustainability, and general operating and behavioural models (Murphy 2012). It's clear that the social and cultural dimensions are closely linked. Cultural processes influence our social lives and how we view social sustainability, like the value we place on equality as a social goal. In turn, social structures and institutions influence cultural practices and judgements. Despite this strong connection, cultural and social sustainability can also be seen as separate dimensions or perspectives of sustainability.

Culture does not have its own Sustainable Development Goal (SDG). In the 2030 Agenda, culture is mentioned in relation to "civilization", "diversity", "interculturality", "cultural heritage", and "tourism", under the following four SDGs: quality education (SDG 4), decent work and economic growth (SDG 8), sustainable cities and communities (SDG 11), and responsible consumption and production (SDG 12) (Duxbury, Kangas, and De Beukelaer 2017).

The concept of cultural sustainability can be interpreted in different ways: as the fourth dimension of sustainability, as culture for sustainability, and as culture as sustainability. The interpretation of "culture as sustainability" requires a new way of thinking and a new paradigm in relation to sustainability. In this view, culture itself is transformed towards sustainability. Cultural sustainability here means thinking about what needs to be preserved, what needs to change, and how we can implement these changes. This interpretation – viewing culture as sustainable development – defines culture in the broadest sense as a comprehensive lifestyle and a continuously changing process. Our social order (such as capitalism and democracy), our values, and our ways of working are all cultural products. Culture thus encompasses all the other dimensions of sustainability and transforming it becomes a key concern or paradigm for sustainability.

The definition of cultural sustainability contains several important aspects: Firstly, it emphasizes the importance of intellectual growth and responsibility. This means acquiring knowledge, developing a better understanding of important issues, and participating in educational activities and events. It also means acting not only as individuals, but also as members of different communities. All problematic and unsustainable structures and operating models have been created and developed by humans. We humans, therefore, also have the opportunity to change them.

Cultural sustainability is required, for example, for a "Great Transformation" as advocated by Uwe Schneidewind (2018). In his work, Schneidewind emphasizes the need for comprehensive change in all dimensions of society, including culture, in order to achieve sustainable development. Recognizing cultural sustainability as a central paradigm of sustainability supports the vision of a comprehensive transformation that goes beyond technological and economic solutions and strives for a fundamental societal change of course.

1.9 Weak and strong sustainability

The academic debate on sustainability distinguishes between weak and strong sustainability (see e.g. Ott 2020). Neither concept can easily be dismissed, as both rely on assumptions that are difficult to refute. In practice, both concepts are considered partially true, with evidence both in favour and against. The key difference between weak and strong sustainability is the question of what we should sustain, and the degree to which what forms of capital can be substituted. This concerns, in particular, the following three forms of capital: natural capital, manufactured capital , and human capital.

Natural capital: the natural resources and ecosystems that are key to human well-being and sustainable development. These include land, water, air, biodiversity, and renewable energy. Natural capital is the basis for many economic activities, and it provides essential ecosystem services such as food, the purification of water and air, and cultural and aesthetic benefits.

Manufactured capital: the material resources and infrastructure used in production and economic activity. This includes buildings, machinery, technological equipment, means of transport, and physical infrastructure. Manufactured capital is closely linked to economic growth and productivity, and plays an important role in economic development.

Human capital: the knowledge, skills, level of education, health, and creative abilities of people in a society. It encompasses individual and collective knowledge as well as the skills needed for economic productivity and social progress. Human capital is important for innovation, labour productivity, social participation, and the ability of a society to tackle challenges and to adapt.

1.9.1 Weak sustainability

Weak sustainability can be considered the "soft" or flexible option. Supporters of weak sustainability believe that an action is sustainable if it offers an advantage to the overall system, or at least does not reduce its quality. Weak sustainability can therefore be seen as a basic requirement of sustainability, in that it aims to maintain a certain level of well-being/quality of life or to ensure equality. The concept of weak sustainability "assumes the extensive and, at least in principle, unlimited […] substitutability of all types of capital" (Ott and Döring 2004, 41) and is thus based on the premise of neoclassical economics. Proponents of weak sustainability believe that natural capital can be substituted with other forms of capital. They emphasize the concept of "total capital", arguing that the specific make-up of the capital inherited by future generations is irrelevant, as long as overall benefits and well-being are sustained. This aligns with neoclassical utility theory, according to which it is irrelevant how utility is generated.

Weak sustainability supporters therefore believe that measures can be considered sustainable, even if they are carried out at the expense of natural capital – as long as this loss is compensated for by an increase in human or manufactured capital. This could justify the extraction of raw materials such as coal or excessive crude oil. And Carlowitz's *Silvicultura oeconomica* forest, could, in principle, also be replaceable or degradable, provided that its natural and cultural functions can be fulfilled in other ways by future generations. This understanding of sustainability emphasizes the importance of technological progress for sustainable development.

Evaluating measures based on weak sustainability is challenging, due to the inherent complexities and numerous assumptions involved. This goes beyond fundamental questions about sustainability, such as "What is a good life?", or "How do we measure the level of well-being?". A key challenge lies in assigning a monetary valuation to different types of capital, especially those lacking a clear market price. To operationalize this, what factors should be taken into account? What is the value of natural beauty? What is the value of a rare bird or a whale? Buller's *The Value of a Whale* (2022) impressively outlines the problems we face in trying to put a price on elements of nature.

Critics of the concept of weak sustainability point to several limitations. Firstly, they question the assumption that natural resources can be endlessly replaced by reproducible capital. Secondly, they doubt whether an increase in goods can truly compensate for the loss of environmental quality. Finally, concerns exist about our ability to develop new resources and the potential for exceeding critical threshold values. These criticisms have paved the way for the concept of "strong sustainability".

1.9.2 Strong sustainability

In contrast to weak sustainability, strong sustainability emphasizes the preservation of natural capital. Proponents of this eco-centric theory are less optimistic about our ability to substitute natural resources with manufactured ones, believing these resource types to have limited interchangeability (H. E. Daly 1990; Ott 2001; Ott and Döring 2004). They argue that the individual elements of natural capital should be kept as constant as possible, to prevent, for example, species extinction.

Table 1.2: Weak and Strong Sustainability. Source: Derived from Eblinghaus & Stickler (1998); Dobson (2002); Rieckmann (2004); Steurer (2001); Michelsen et al. (2016).

Category	Very Weak Sustainability	Weak Sustainability	Strong Sustainability	Very Strong Sustainability
What should be preserved?	Total capital	Essential natural capital	Non-renewable natural capital	Nature has its own value
Why?	Maximization of economic profit and individual welfare (GDP)	+ Limitation of environmental damage and resource scarcity	Ensuring long-term living conditions for humans and nature	+ Respect for all forms of life and duties toward nature
How? (Policies)	Growth orientation through promotion of technology and innovation	Environmental regulations, resource efficiency, renewable energies	+ Conservation measures, ecological restoration, sustainable agriculture	
Substitutability	In principle unlimited, natural resources can be replaced by technology and trade	Partly possible, but limited substitutability of ecosystem services	Limited, critical threshold for environmental changes	Low, recognition of unique values and relationships
Ethics	Anthropocentric, short-term benefits take priority	Anthropocentric with stronger consideration of environmental interests	Pathocentrism, recognition of intrinsic value of nature	Biocentrism, ecological integrity, ethical responsibility toward nature

While neoclassical economists are confident about substitution, proponents of strong sustainability emphasize the importance of prevention and anticipation, rather than aftercare and reaction. For example, reacting to the hole in the ozone layer by using sun cream, protective clothing, or medical aftercare fails to tackle the cause of the problem. Similarly, technological approaches such as geo-engineering are criticized for merely treating problems superficially. Geo-engineering refers to technical interventions in geochemical or biogeochemical cycles, for example to slow down climate change or ocean acidification.

Until now, reducing pollution has mainly focused on end-of-pipe technologies, such as flue gas cleaning systems on factory chimneys or catalytic converters in cars. But this also meant that we put off tackling long-term environmental problems, such as climate change or biodiversity loss, as long as their effects were not immediately apparent. This phenomenon is linked to the concept of externalization, in which environmental and social costs are shifted outwards, as described by Lessenich (2018). This means that pricing only takes into account the economic costs, partly because they are easier to quantify, while the social and environmental costs of providing goods and services are omitted or externalized, leading to distortions in market prices.

1.10 Sustainability strategies as principles of sustainable development

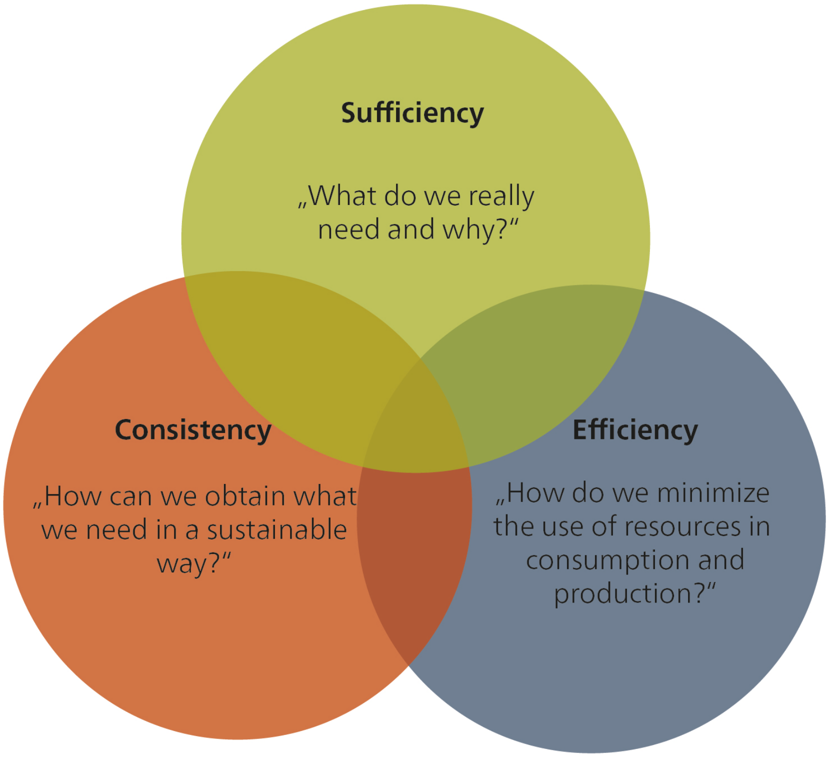

Figure 1.12: Sustainability strategies. Source: Own illustration.

1.10.1 Efficiency strategy

The efficiency strategy focuses on minimizing resource use (i.e. raw materials and energy) in manufacturing or service provision. This translates to reducing material consumption (material intensity), energy consumption (energy intensity), and emissions of harmful substances such as CO_2. Also known as “eco-efficiency”, this strategy is attractive to business and society, as it can reduce costs, resource use, and environmental impact. Implementing the efficiency strategy starts with improving production processes, primarily through technological advancements. Proponents believe it’s possible to double prosperity while halving natural resource use (von Weizsäcker, Lovins, and Lovins 1995). However, critics of the efficiency strategy are less optimistic, and caution against overestimating its impact on sustainable economic activity. They point to “rebound effects”, which can reduce or even wipe out the gains from efficiency increases (Paech 2012; Santarius 2014).

Putting Efficiency Strategies into Action: Examples

Energy-efficient lighting: Replacing conventional lightbulbs with energy-efficient LED bulbs lowers electricity use and helps reduce the demand for energy.
Fuel-efficient vehicles: The development of more fuel-efficient hybrid or electric vehicles can lead to lower fuel consumption and road transport emissions.
Efficient building technology: Installing smart HVAC (heating, ventilation, and air conditioning) systems in buildings helps to reduce energy consumption for heating and cooling.
Efficient water use: The use of water-saving fittings and systems in households and industrial companies reduces water consumption and minimizes waste.

1.10.2 Consistency strategy

While the efficiency strategy focuses on quantity (reducing resource use while achieving greater output), the consistency strategy (from the Latin con = together + sistere = to stop) strives to reconcile nature and technology by using resources repeatedly instead of only once. Instead of fossil fuel-based resources, products, and technologies, it seeks to use materials that are compatible with natural material cycles and processes. Also referred to as “eco-effectiveness”, this concept follows the principle of “cradle to cradle” rather than “cradle to grave”, and is based on the idea that intelligent systems generate only products, not waste. This can be achieved in two ways: Materials can either be biodegradable (e.g. a shampoo made with natural ingredients) or they can be designed with “technical nutrients” that remain in the technical cycle. This means that a disused product doesn’t end up in the rubbish, but instead enters the next cycle of use, for example through upcycling (e.g. reusing a computer casing or converting into, say, a shelving system). Like the efficiency strategy, the consistency strategy also starts by looking at how production processes can be optimized. Many believe that the consistency strategy has greater potential than the efficiency strategy, in terms of problem solving and achieving far-reaching changes.

However, critics believe the theory has limited applicability. For example, Gerolf Hanke and Benjamin Best (2013) argue that it can’t be applied equally to every type of good, and that implementing technologies to achieve it would require significant investment in production facilities and logistics. They also note that recycling processes themselves are associated with increased energy consumption, in accordance with the law of entropy:

> “Every material economic process results in an increase in entropy, i.e. roughly simplified, that the elements on the material level are distributed more and more evenly, which ultimately only means that things and bodies wear out and a new concentration requires an ever-increasing amount of energy.” (Translated from Hanke and Best 2013, 11)

A special feature of the Earth's ecosystem is solar energy, which provides a constant supply of energy from the outside and counteracts the increase in entropy on Earth. But we still need what is known as "grey energy" to manufacture energy generation systems, e.g. biofuel systems, solar cells, electric car batteries, and wind turbines. Consistency strategies can incentivize industry to strive for a reduction in resource consumption and emissions.

Putting Consistency Strategies into Action: Examples

Renewable energy systems: The transition from fossil fuels to renewable energy sources such as solar energy, wind energy, and hydropower is an effort to align energy production with the natural rhythms and cycles of the environment.
Recyclable electronics: Designing electronic devices in a way that makes it easy to repair, upgrade and recycle them, in order to minimize resource consumption and environmental impact.
Sustainable construction: Constructing buildings using sustainable materials that have a low environmental impact and can be reused or recycled at the end of their useful life.

1.10.3 Sufficiency strategy

The concept of sufficiency, rooted in the Latin word sufficere (to suffice), challenges our assumptions about how much we need for a good life. It promotes a reduction in resource and energy consumption by focusing on decreasing demand for resource-intensive goods and services. Sufficiency, which could also be described as frugality or adequacy, is critical of the constant creation of new needs by technology and advertising amid a limited supply of natural resources. Sufficiency urges us not to chase after every newly created need, and to fulfil our needs without consumption. While efficiency and consistency strategies focus on production, sufficiency focuses on consumption, albeit not exclusively: sufficiency can be practised to varying degrees and at different levels, from small changes in behaviour (sharing instead of buying) to significant changes in lifestyle (giving up air travel). Although it starts at an individual level, sufficiency can be applied at various levels, such as by companies (sufficiency-oriented product design) and governments (sufficiency policies). Sufficiency therefore seeks the right balance: How much do we need for a good life? And what do we not need?

Critics believe the sufficiency strategy has only limited potential and is unlikely to find broad sociocultural acceptance. However, proponents see it as crucial for sustainability policy, especially where efficiency and consistency strategies fall short. The concept of a post-growth economy draws on the principles of the sufficiency strategy.

Putting Sufficiency Strategies into Action: Examples

Sharing and communal use: Platforms and initiatives for sharing objects, tools, or vehicles make it possible to use resources more efficiently and reduce the number of products manufactured.
Plant-based diet: Switching to a predominantly plant-based diet reduces resource consumption, e.g. of water and land, compared to meat production.

Reduction in working hours: A reduction in working hours can lead to a lower consumption of resources, as less energy and materials are required for the production of goods and services.
Local consumption: Supporting local producers and markets helps to reduce transport routes and emissions.

1.10.4 Rebound effects: The flipside of efficiency

Figure 1.13: Rebound effect illustrated using the example of the automotive industry. Source: @fietsprofessor on LinkedIn.

"Any intelligent fool can make things bigger, more complex, and more violent. It takes a touch of genius — and a lot of courage to move in the opposite direction." (Schumacher 1973)

Critics of the efficiency principle warn of the rebound effect, also known as the Jevons paradox (Jevons 1865/1965) . This phenomenon occurs when efficiency improvements lead to lower prices for goods and services, thus increasing demand and negating the initial benefits of lower resource use.

Take this simple example: If passenger cars become cheaper due to improvements in efficiency, then people may choose a larger model when they next purchase a car. In addition, a fuel-efficient car is cheaper to run, as it requires less fuel per kilometre. This usually incentivizes people to drive more (either by making more frequent journeys or travelling longer distances) and reduce their use of public transport or their bicycle. As a result, efficiency gains that are technically possible are often not realized in practice, as the product in question is used more frequently or more intensively. Apart from the direct change in the use of the product (direct rebound), there may be additional changes in consumer behaviour that affect the environment. In this example, this could mean that the money saved on purchasing or using the car is spent on air travel instead (indirect rebound), which in turn cancels out some of the environmental benefits.

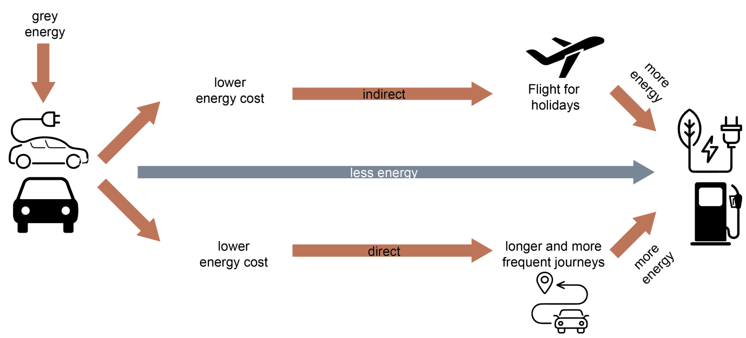

Figure 1.14: Indirect and direct rebound effects of energy-saving passenger cars. Source: Own illustration.

To understand the rebound effect, the literature focuses on the following three areas: 1) financial factors, 2) socio-psychological influences, and 3) regulatory effects, which we analyse in more detail below.

1. Financial factors are a major cause of the rebound effect. When efficiency leads to cost savings, financial resources are freed up. This can lead to people either consuming more of the same product (direct rebound effect) or investing in alternative goods (indirect rebound effect). The financial rebound effect can be based on several factors, including:
   - The income effect: when efficiency leads to real income gains for consumers, it can result in a direct or indirect rebound effect.
   - The reinvestment effect: when companies use cost savings to expand their production or invest in other products.
   - The market price effect: the macroeconomic effect in which demand in one sector stimulates demand in other sectors. For example, as cars become more efficient, demand for fuel may fall, making fuel cheaper and affecting demand in other energy-consuming sectors.

The extent and intensity of the financial rebound effect depends on factors such as price elasticity and consumer behaviour. While this happens at the individual level, it can cumulate into a macroeconomic market price effect with far-reaching consequences.

2. Socio-psychological influences include the “mental accounting” of individual consumers. For example, consumers who have purchased an efficient product may “allow themselves” to consume more. This is why increases in the efficiency of products that were previously considered harmful to the environment can, paradoxically, lead to an increase in consumption. This direct rebound effect is often referred to in social psychology as the “moral hazard effect”. If the decision to increase consumption is not made rationally, this is known as the “moral leaking effect”. For example, if you drive more after buying a resource-efficient car, because the purchase alone is perceived as environmentally conscious. The

indirect rebound effect can also be attributed to socio-psychological factors. The "moral licensing effect" states that the purchase of resource-efficient products can lead to an increase in the consumption of other products that may be harmful to the environment. For example, when the purchase of a fuel-efficient car is used to justify more frequent air travel.

3. Sometimes, government regulations or incentives to promote efficient technologies can have unintended consequences – this is what we mean by the "regulatory effect". For example, if people are encouraged to purchase large electrical appliances devices due to government incentives for energy-efficient models. Such incentives may result in more frequent purchases of appliances, which – even if the newer model is more efficient than the old – partially cancels out the energy-savings effect. Furthermore, the introduction of efficiency technologies leads to new capacities and infrastructures, which can result in the creation of new markets. Wind turbines, for example, promote the development of new infrastructures and jobs, but at the same time require considerable resources. In addition, consumers who purchase a more efficient product may not necessarily discard their old product. Some end up using both products, which can lead to increased overall consumption (Santarius 2014).

Rebound-effects

The **direct rebound effect** occurs when improved energy efficiency makes it cheaper to consume a resource, leading people to consume more of it. For example, a more fuel-efficient car would be cheaper to run. This might encourage people to drive more, partially or completely cancelling out the energy savings from the efficiency increase.
The **indirect rebound effect** occurs when energy savings or resource efficiency gains in one area lead to the use of the freed-up resource in other area. This reduces the overall savings achieved. For example, if the money saved from more energy-efficient living is used instead for more frequent flying.

1.10.5 How big is the rebound effect?

It is difficult to know exactly how big the rebound effect is: empirical estimates vary widely, depending on the methods used and the effects taken into account. It is particularly challenging to clearly distinguish rebound effects from growth or structural changes. The type of products and services offered also influences the level of the rebound effect. For example: If a more economical means of transport is used for the daily commute, such as a more efficient car, costs fall, but the available time remains limited. Therefore, you cannot commute as often or for as long as you like, because there are only 24 hours in a day. The rebound effect is therefore relatively small in this case. The situation is different for leisure activities, such as travelling by air. If costs fall due to more efficient aircraft or cheaper prices, there is a greater incentive to fly more often — for example, taking a second weekend trip instead of just one per year. As time is not such a tight constraint here, additional consumption can increase significantly — as can the rebound effect. Another important component is the degree of saturation with goods and services. Observations show that rebound effects tend to be lower in high-income countries than in developing countries, where there is still a considerable need for additional consumption.

For example, the direct rebound effects associated with improvements in space heating efficiency might lead to 10-30% less energy savings than what's technically possible (e.g. Hediger, Farsi, and Weber 2018). The rebound effects of transport vary even more. According to Andersson, Linscott, and Nässén (2019), the rebound effect related to private mobility might reduce potential savings by 7.5% in Denmark, 30% in Sweden, and 60% in Germany. In contrast, studies on lighting in private households have found very low rebound effects of less than 10% (Sorrell 2007). This means that the actual energy savings for these services can be, on average, up to 25% less than what is technically possible and predicted. However, the exact magnitude of the rebound depends on the specific context and can be reduced by choosing appropriate measures.

Sometimes, albeit rarely, the savings may be overcompensated. This is known as the "backfire" effect. However, this is an exception and the "backfire" effect is not a pure rebound effect, as it is linked to the effects of growth and structural change. A good example is digitalization, which can lead to short- and medium-term backfire effects (Peng, Zhang, and Liu 2023).

1.11 Sustainability as a scientific discipline

What does "sustainable" mean? We could argue that sustainability means the existence and functioning – for as long as possible – of structures of various degrees of complexity. There are, in fact, many structures in the world that have existed for relatively long periods of time, such as atoms, certain ecosystems, life on Earth as a whole, or planetary systems and galaxies. But why are some structures relatively stable and able to exist "sustainably", despite being exposed to regular disturbances that jeopardize their existence or function? Are there principles or mechanisms that could explain the continued existence of these diverse systems? And if so – could human societies learn from these principles?

As early as the mid-20th century, scientists began to recognize that fundamental principles connect all aspects of existence, shaping the universe's order and dynamics. A key figure in this field was Karl Ludwig von Bertalanffy (1901–1972). His research began with his reflections on the "formation of form" in nature (1928), which he further developed in publications on the structure of life (1937), molecules and organisms (1940), and biology (1949). In 1950, he introduced a theory of open systems in physics and biology, a concept that significantly influenced the development of biophysics, fluid equilibrium, and modern developmental theories. His seminal work, "General System Theory", was published in 1968.

Systems theory laid the foundation for a revolutionary world view that not only provided explanations for the development and function of so-called systems, but also integrated isolated knowledge from various disciplines. In 1975, in a significant contribution to systems theory, Ervin Laszlo distinguished between seven different system types (physico-chemical and biological systems, organ systems, socio-ecological and socio-cultural systems, and organizational and technical systems). Today, systems theory forms the basis for models for the development of complex systems and for cybernetics, which deals with the control of systems. The Club of Rome report used systems theory to analyse the growth of processes in the context of positive feedback.

1.11.1 Systemic effects: Escalation and feedback

In systems theory, the mechanisms of positive and negative feedback are used to explain the dynamics and stability of systems. Positive feedback reinforces changes in a system, while negative feedback tends to balance changes, returning the system to a stable state.

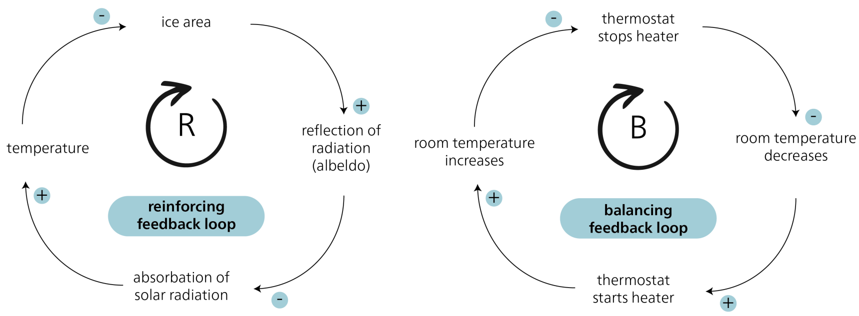

Figure 1.15: Positive/reinforcing and negative/balancing feedback loops. Source: Own illustration.

An example of **positive feedback** in relation to the accelerated melting of Artic Sea ice is what is known as albedo amplification. The Arctic ice reflects much of the solar radiation back into space, resulting in less absorption of heat by the ocean. However, when the ice melts, less sunlight is reflected, as the seawater, which is darker, absorbs more heat. This warms the seawater further and the remaining ice melts more rapidly. This process is self-reinforcing, as less ice leads to a lower albedo, which in turn leads to greater absorption of sunlight and further melting. This positive feedback has serious consequences on the climate system, as the dwindling Arctic Sea ice can lead to a faster sea level rise and to changes in oceanic and atmospheric circulation patterns. The extent and speed of this occurrence took scientists by surprise (Stroeve et al. (2007)).

An example of **negative feedback** is a thermostat that regulates the temperature in a room. If the room temperature rises above the set maximum, the thermostat recognizes this and activates the air conditioning to lower the temperature. As soon as the temperature reaches the set value, the thermostat switches off the air conditioning. This control circuit ensures that the room temperature is maintained at a constant level.

Positive and negative feedback often work together and can cause complex behaviour in systems. Understanding these feedback mechanisms makes it possible to better understand and predict the interactions and behaviour of systems.

1.12 Sustainability science: origins and understanding

An emerging awareness – influenced not least by the publication, *The Limits to Growth* (Meadows et al. 1972) – led to increased research into global environmental problems. Three major international research programmes are particularly noteworthy here: 1) the *International Geosphere-Biosphere Programme* (IGBP) from 1987 to 2015; 2) the *World Climate Research Programme* (WCRP), founded in 1980; and 3) the International Research Programme on Biodiversity (DIVERSITAS), founded in 1991 and renamed in 2014 to *Future Earth*. Not only did these programmes initiate important basic research – they also provided an important impetus for greater consideration of the results in policy, as reflected, for example, in the UN's Brundtland Report and Agenda 21. Notably, these research programmes systematically brought together previously separate disciplines of the natural sciences for the first time, although they still largely neglected the interaction between society and the environment. As the interconnectedness of these topics began to gain traction, several research projects promoting their integration were launched, particularly in the US.

> "A new field of sustainability science is emerging that seeks to understand the fundamental character of interactions between nature and society. Such an understanding must encompass the interaction of global processes with the ecological and social characteristics of particular places and sectors."(Kates et al. 2001, 641)

The concept of sustainability science was officially introduced in 2001 at the "Challenges of a Changing Earth" congress in Amsterdam, by the International Council for Science (ICSU), the above-mentioned IGBP, the International Human Dimensions Programme on Global Environmental Change (IHDP), and the above-mentioned WCRP. The German-language term "Nachhaltigkeitswissenschaft" (sustainability science) can be traced back to a translation from English.

1.12.1 Understanding sustainability science

An oft used motto – "society has problems – universities have disciplines" – epitomizes the conventional structure of science into various disciplines. However, disciplinary science appears to be inadequately equipped to meet the challenges of sustainable development. Two main approaches to understanding sustainability science have emerged: on the one hand, it is seen as an independent discipline with its own theories and methods; on the other, as an amalgamation of different disciplines focusing on a common theme (Clark and Dickson 2003). While the first approach is not yet considered accurate, there is broad agreement that sustainability science should be seen as a platform that brings together science, practice, and visions – and incorporates contributions from the entire spectrum of natural, economic, and social sciences (Martens 2006, 38).

Sustainability science is distinct in two key ways. First, unlike traditional ("Mode 1") research, it isn't driven by an original scientific programme. Instead, it is guided by the normative concept of sustainability, using this as a framework for scientific analysis. Second, as a distinct type of problem-oriented research, it differs from both basic and applied research. Clark (2007) introduced the term "use-inspired basic research", which seeks to strike a balance by seeking new knowledge while also considering its potential applications .

Table 1.3: Mode-1 and Mode-2 science. Source: Based on Gibbons et al. (2010) and Martens (2006).

Mode-1 science	**Mode-2 science**
Academic	Academic and social
Mono-disciplinary	Trans- and interdisciplinary
Technocratic	Participative
Certain	Uncertain
Predictive	Exploratory

1.12.2 Empirical-analytical vs normative

As sustainable development is a normative model, it is important to distinguish between descriptive (empirical-analytical) research and prescriptive (normative) research. Descriptive statements describe the current state ("is"), while normative statements describe the ideal state ("ought"). In everyday life, the distinction between "is" and "ought" is often not recognized, but it is crucial for science.

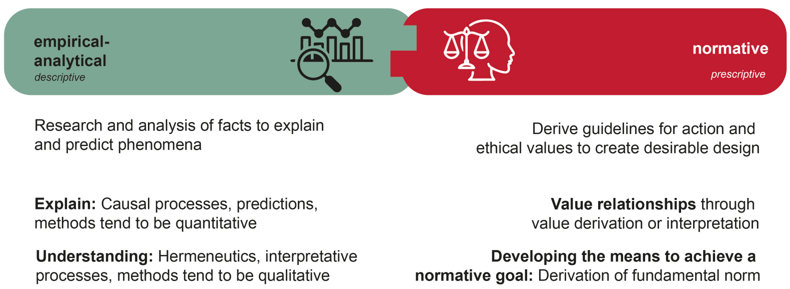

Figure 1.16: Distinction between empirical analytical and normative. Source: Own illustration.

It is important to note that "is" and "ought" statements have different qualities and focus areas, and they cannot simply be linked without further explanation. This understanding forms the basis of Hume's law, which declares that normative statements about "what ought to be" cannot be derived from descriptive statements about "what is". To give an extreme example: The fact that around 24,000 people starve to death every day does not allow any conclusions to be drawn as to whether this is good or bad. Without an evaluation, no normative statement can be made about whether this should be prevented. Hume pointed out that many scientists of his time often neglected this crucial distinction, and this is still the case today:

> "In every system of morality, which I have hitherto met with, I have always remark'd, that the author proceeds for some time in the ordinary ways of reasoning, and establishes the being of a God, or makes observations concerning human affairs; when of a sudden I am surpriz'd to find, that instead of the usual copulations of propositions, is, and is not, I meet with no proposition that is not connected with an ought, or an ought not. This change is imperceptible; but is however, of the last consequence. For as this ought, or ought not, expresses some new relation or affirmation, 'tis necessary that it shou'd be observ'd and explain'd; and at the same time that a reason should be given; for what seems altogether inconceivable, how this new relation can be a deduction from others, which are entirely different from it… [I] am persuaded, that a small attention wou'd subvert all the vulgar systems of morality, and let us see, that the distinction of vice and virtue is not founded merely on the relations of objects, nor is perceiv'd by reason." (Hume 1739, bk. 3, Part 1, Section 1)

The realization that is and ought statements differ does not mean that scientists cannot make ought statements. Rather, it means that due to the epistemological differences between the two categories, certain rules must apply when they are linked. Hume tried to establish the relationship between these two statement types through so-called "bridge principles". Max Weber further formalized these considerations by replacing the bridge principles with the concept of "value relations". Weber's call for value freedom (or value neutrality) in empirical science stems from his observation that there is a clear distinction between factual questions ("is") and value questions ("ought") (Weber 1999, p. 245f, 509f). He argued that it is not the task of the empirical sciences to make value judgements. If scientists fail to make the analytical distinction between their personal values and their research goals, this could jeopardize the ability of science to produce verifiable and meaningful results (Weber 1999, p. 150f; Strauss 1953). In turn, this could lead to a blurring of the boundaries between scientific knowledge and knowledge from politics, religion, or everyday discussions.

Weber's call for value freedom in empirical science was often misunderstood (Jonas 1980, p.189f; Weber 1999, p. 499f). It does not mean that research should do without values. (Non-scientific) values have a place in empirical research, particularly in the context of sustainable development, in the following ways:

- **Values can be an object of inquiry (research object):** Values play a key role in the field of environmental and sustainability psychology, for example. In discussions about lifestyles and subjective well-being, researchers study the personal values and beliefs of individuals.

- **Values influence the choice of research topic:** Values influence the selection of research objects and questions (epistemological interest), and thus the direction of research. This is particularly relevant in application-oriented and transdisciplinary research, where the needs and information of relevant social actors are integrated into the research process, for example through *joint problem framing* or *participatory vision development* (Schneider et al. 2019).

- **Value interpretations should be clearly stated:** Researchers can create value relationships, or make value interpretations (cf. Weber 1999, p. 522). A value interpretation, based on the hermeneutic method, concretizes the meaning of overarching value concepts and derives logical values and norms from them. For example, researchers can analyse the SDGs by examining the values underlying sustainable development. Other examples include norms derived from the concept of justice – such as the norm of human responsibility towards our environment, our social environment, and ourselves (see Michelsen and Adomßent 2014, 25; Schneider et al. 2019). It is important to make the original value explicit.

- **Developing the means to achieve a normative goal:** Sustainable development research often relies on value judgements to guide its pursuit of evidence-based solutions. This is comparable to political science, where normative theories aim to construct an optimal – i.e. just – form of government, based on the positive evaluation of what constitutes a just society. Science cannot prove the value of justice, but it can develop strategies and means to achieve this desired goal. This relationship between ends and means represents a logical connection between values and normative statements. The ultimate goal, or end, is determined by extra-scientific values. The researchers have to clearly state this relationship, and formulate their means accordingly. For example, an end–means connection could look like this: "To achieve goal X, Y is the most successful means under conditions b1, b2, and b3". Sustainable development can be considered as an extra-scientifically given fundamental norm. For example, if research wants to make statements about how sustainable development can be achieved through solar energy, it could proceed as follows: "Switching to solar production could be a promising way of achieving sustainable development. This would organize production and distribution according to the needs of many people, minimize social inequalities, and create conditions for individual well-being."

1.13 Types of knowledge for sustainable development

Sustainability research focuses on challenges that threaten the long-term safeguarding of social development conditions. In the context of SD as a challenge for society as a whole, the Forum for Climate and Global Change of the Swiss Academy of Sciences has described three different types of knowledge (ProClim 1997): systems, target, and transformation. These three knowledge types take place at three fundamental levels: 1) the analytical level, which aims to generate systems knowledge; 2) the normative level, which develops target and orientation knowledge; and 3) the operational level, which seeks to generate design or transformation knowledge. Figure **??** shows the three knowledge types in relation to empirical-analytical and normative research.

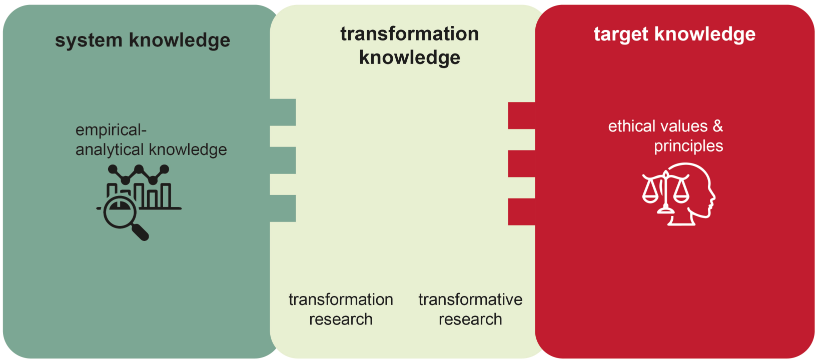

Figure 1.17: The three types of knowledge. Source: Own illustration based on ProClim (1997).

Systems knowledge refers to the understanding of the current state. In relation to sustainable development, research examines relevant structures and processes, describing and explaining their complexity and dynamics. It uses various methods, including qualitative and quantitative approaches, mixed methods, experiments, and systems dynamics approaches. It also uses forecasts, retrospectives, and scenario analyses. Most of the studies that generate systems knowledge fall within the field of empirical research.

Transformation knowledge is the knowledge of how we can get from the current state ("is") to the desired state ("ought"). This requires an integration of both systems and target knowledge. In other words, research is looking for demonstrably effective solutions to the problems identified in the context of sustainable development. There are many methods to generate transformation knowledge.

Research on transformations, i.e. the change in social organization at various levels, can be empirical as well as normative. Research focusing on past and present transformations is empirical, while research focusing on concrete goals is normative. Accordingly, the German Advisory Council on Global Change (WBGU), distinguishes between "transformation research" and "transformative research".

Another concept that is emphasized in the context of transformations is "transformative education", which seeks to promote skills for a comprehensive and multi-layered understanding of change (Schneidewind 2013). As such, transformative education aims to strengthen society's ability to reflect on issues, for example in terms of observation and in actively shaping transformation processes.

Target knowledge refers to the knowledge of what should be achieved and what should be avoided. Normative research is particularly important in this context. It is about interpreting and concretizing the values of sustainable development and deriving further standards. Research into the definition of threshold values is just as relevant as methods for assessments and visioning processes (cf. Swart, Raskin, and Robinson 2004; Wiek and Iwaniec 2014). Target knowledge gives us the answer to the question of what we should transform. CDE's study programmes are primarily guided by the following concepts of target knowledge for a sustainable economic and social system:

- The principles of sustainable development
- The 2030 Agenda and the SDGs (2015)
- The WBGU's Flagship Report "World in Transition: A Social Contract for Sustainability" (2011)
- The Global Sustainable Development Report GSDR (2019)

Part II

Approaches to Sustainability

Chapter 2

Approaches to Sustainable Development

The world is facing numerous challenges of sustainable development, including pressing environmental problems and social inequalities. Scientists, researchers, and activists are seeking innovative approaches to enable sustainable and equitable change. This chapter examines some of these approaches, which offer a complete rethink of current paradigms.

Approaches discussed in this chapter:

- Planetary boundaries framework
- Doughnut economics
- Approaches to a "great transformation"
- Green economy
- Post-growth approaches
- Implementing the 2030 Agenda

2.1 Debates about planetary boundaries

In 1798, British economist Thomas Robert Malthus published his influential essay, *An Essay on the Principle of Population*. Malthus's core idea was that population growth would surpass Earth's ability to produce food, sparking a debate on planetary carrying capacity that continues today. *The Limits to Growth* (1972), a report by the Club of Rome, built on Malthus's ideas. It went beyond just food supply to consider a broader range of factors in calculating a planetary limit. These factors included resource availability, environmental pollution, and industrial output. The report remains relevant from an environmental point of view, as it challenges the assumptions of limitless growth, even though its specific predictions haven't quite come true.

A significant development in sustainability science is the planetary boundaries concept, introduced in 2009. Unlike earlier theories focused solely on population limits, this framework uses Earth system science parameters. Researchers identified the Holocene epoch as a baseline, as this was a period of remarkable stability for human development. Significant deviations from this ideal state could push humanity towards uncertain "tipping points" – critical thresholds that can either interrupt previous progress, alter its course, or even accelerate it in unintended ways. An example might be the extinction of megafauna at the end of the last ice age, potentially linked to human arrival in the Americas. The planetary boundaries concept promotes the precautionary principle, urging action to minimize potential harm to both humans and the environment.

The concept, which puts forward nine planetary boundaries, was first introduced by Johan Rockström et al. (2009). It was updated by Will Steffen et al. (2015). A recent update by Richardson et al. (2023) shows that six out of nine planetary boundaries have already been transgressed.

The planetary boundaries framework identifies nine critical Earth system processes. These processes regulate the planet's stability and include, for example, land system change and ocean acidification (see all planetary boundaries here: Figure **??**). Each boundary has an inner circle, within which it can operate safely ("safe operating space") and an outer circle, which represents increased uncertainty.

The latest update to the planetary boundaries framework paints a concerning picture. We're close to overstepping the safe operating space for ocean acidification, and regional atmospheric aerosol loading has already crossed its boundary. In a positive development, stratospheric ozone levels show some signs of recovery. However, the overall situation is alarming. The boundaries previously identified as transgressed (climate change, biosphere integrity (genetic diversity), land system change, and biogeochemical flows [N and P]) have all seen a worsening of their transgression since 2015.

The study added human appropriation of net primary production as a control variable for the functional component of biosphere integrity, arguing that this boundary has also been exceeded. In addition, the significant transgression of the planetary boundaries for phosphorus and nitrogen cycles, along with genetic biodiversity, raise the risk of fatal consequences.

Two of the nine planetary boundaries – biosphere integrity and climate change – are considered "core boundaries". These core systems encompass processes from many other subsystems and operate at a global scale. Reaching tipping points in these core systems could therefore push the entire Earth system into a new state.

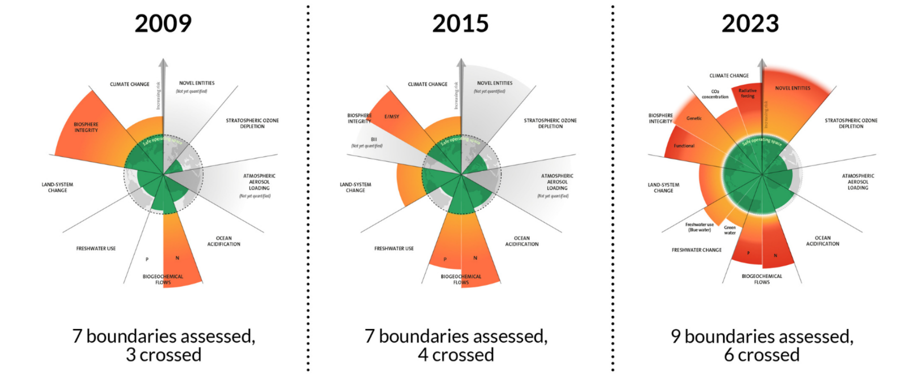

Figure 2.1: Development of control variables for all nine planetary boundaries. Source: Azote for Stockholm Resilience Centre, Stockholm University. Based on Rockström et al. (2009), Steffen et al. (2015), Richardson et al. (2023). Licenced under CC BY-NC-ND 3.0.

2.1.1 Tipping points of the Earth's climate system

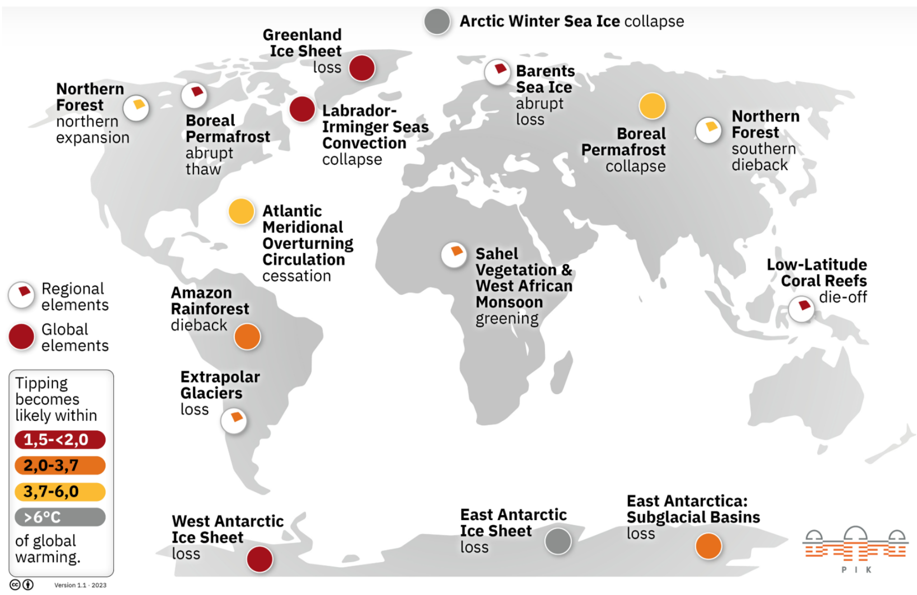

Figure 2.2: Spatial distribution of global and regional tipping elements. The colours indicate the temperature range in which tipping is likely to occur. Source: Designed at PIK, based on Armstrong McKay et al. (2022), expanded with interactions based on Lenton et al. (2019). Licenced under CC-BY.

Tipping points in the Earth's climate system are critical thresholds that, when crossed, can cause abrupt and often irreversible changes in the climate system. These tipping points can destabilize the climate and lead to accelerated climate change. Examples of tipping points include the melting of the Greenland ice sheet, the collapse of the Amazon rainforest, the thawing of permafrost soils, and changes in the Gulf Stream. If these tipping points are reached or exceeded, they can trigger self-reinforcing feedback effects that lead to further warming and an intensification of climate change.

The concept of tipping points emphasizes the urgency of limiting global warming and reducing greenhouse gas emissions. If we exceed the tipping points, it will become increasingly difficult to control climate change and minimize its effects.

2.1.2 Quantifying planetary boundaries

A recent development within the planetary boundaries framework is the concept of "safe and just Earth system boundaries (ESBs)" for the following domains: climate, the biosphere, water and nutrient cycles, and aerosols at global and subglobal scales (Rockström et al. 2023). The ESBs are based on modelling and literature review, and account for uncertainty through different levels of likelihood. Staying within the ESBs protects stability and equity between species and future generations, although current generations, especially vulnerable groups, could still suffer harm. The authors therefore suggest stricter boundaries in some cases, and the addition of local standards to protect current generations and ecosystems. For example, they identify safe ESBs for warming (Rockström et al. 2023, fig. 1 and Table 1). These are based on reducing the probability of triggering climate tipping points, maintaining biosphere and cryosphere functions, and considering climate variability of the Holocene (<0.5-1.0°C) and earlier interglacial periods (<1.5-2°C).

The functions of the cryosphere include the preservation of permafrost in the northern high latitudes, the preservation of polar ice sheets and mountain glaciers, and the minimization of sea ice loss. The authors conclude that global warming of more than 1.0°C above pre-industrial levels, which has already been exceeded (IPPC 2021), could trigger tipping effects such as the collapse of the Greenland ice sheet or a localized abrupt thawing of the boreal permafrost with a moderate probability (Armstrong McKay et al. 2022). Global warming of one degree Celsius corresponds to the safe limit proposed in 1990 and the PB of 350 ppm CO_2 (Steffen et al. 2015). With a warming of more than 1.5°C or 2.0°C, the likelihood of triggering tipping points increases to high or very high.

2.1.3 Climate resilience

Resilience describes the ability of a system to withstand disruptions, "bounce back", or recover from adversity. Originally used in psychology, resilience refers to the psychological robustness that an individual has actively acquired in dealing with challenges or stresses, particularly in childhood. In the context of ecosystems, resilience refers to the ability to absorb disturbances without a permanent systemic collapse, i.e. a collapse that would result in a different system regulated by new processes (Folke et al. 2010). More recently, the concept of resilience has been extended to social systems (see section on doughnut economics). Studies focus on which specific characteristics of a region need to be strengthened, to better prepare it for future crises and disasters related to climate change, terrorism, resource scarcity, or financial crises. The climate crisis, for example, requires both adaptation and mitigation measures. Resilience approaches offer a way of combining these two concepts rather than playing them off against each other.

Climate mitigation measures aim to reduce greenhouse gas emissions and curb climate change. Such measures include promoting renewable energies, improving energy efficiency, and expanding public transport. Resilience approaches emphasize the importance of climate mitigation, as limiting the rise in temperature will help to reduce the intensity and frequency of extreme weather events.

Climate adaptation measures aim to make societies and ecosystems more resilient to the current and expected effects of climate change. Adaptation measures include the development of early warning systems for extreme weather events, coastal protection against rising sea levels, the reduction of heat stress (e.g. through more urban green spaces), runoff or infiltration areas to reduce the damaging effects of heavy rainfall events, or the adaptation of agricultural practices to changing climatic conditions. Resilience approaches emphasize the need for climate adaptation to protect communities and ecosystems from the negative effects of climate change.

2.1.4 Conclusion

The concept of planetary boundaries tries to reduce complex ecological relationships to a small number of quantifiable limits. These specific limits and indicators for planetary boundaries are based on scientific findings that are not always clear or consistent. The boundaries are therefore contested by some scholars, who question the accuracy and reliability of the data and models used. Despite this criticism, the planetary boundaries framework makes a valuable contribution to the debate on sustainable development and raises awareness of the limited resources and resilience of our planet. To summarize:

- The planetary boundaries framework focuses on the ecological/biophysical limits of the Earth's resilience, and thus the environmental dimension of sustainable development.
- These limits to resilience – the planetary boundaries – focus on environmental factors that are considered fundamental to human survival.
- In normative terms, the framework aims to maintain the stable Earth system ("state of equilibrium") of the Holocene, thereby mitigating threats to human survival.
- The planetary boundaries framework can be used to set concrete targets in the environmental dimension (e.g. at the global or national level).

Further readings

Folke, Carl, Stephen R. Carpenter, Brian Walker, Marten Scheffer, Terry Chapin, and Johan Rockström. 2010. 'Resilience Thinking: Integrating Resilience, Adaptability and Transformability'. *Ecology and Society* 15 (4). https://www.jstor.org/stable/26268226.

Lenton, Timothy M., Johan Rockström, Owen Gaffney, et al. 2019. 'Climate Tipping Points — Too Risky to Bet Against'. *Nature* 575 (7784): 7784. https:

//doi.org/10.1038/d41586-019-03595-0.
Rockström, Johan, Joyeeta Gupta, Dahe Qin, et al. 2023. 'Safe and Just Earth System Boundaries'. *Nature* 619 (7968): 102–11. https://doi.org/10.1038/s41586-023-06083-8.

2.2 From planetary boundaries to the doughnut model

Economist Kate Raworth's Doughnut Model is an innovative approach to sustainable development that recognizes social and environmental limits. This requires rejecting much of what has characterized 20th-century economics, as Raworth outlines in her 2017 book, *Doughnut Economics: Seven Ways to Think Like a 21st Century Economist*. Raworth depicts the ideal economy of the future in a simple image: a ring-shaped doughnut. The outer ring symbolizes an ecological ceiling that should not be crossed, as doing so would cause irreversible harm to the environment. The inner ring represents a social foundation covering people's basic needs, such as food, housing, and income. The challenge is to ensure that economic activities take place within this ring – the doughnut – to ensure the well-being of both humanity and the environment ("safe and just space for humanity").

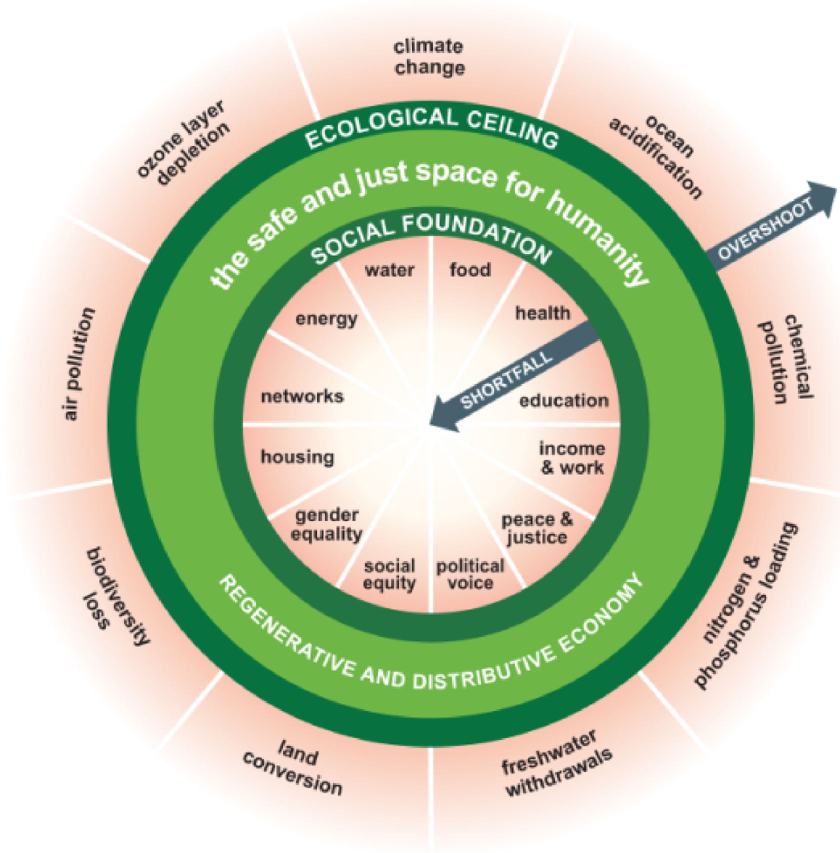

Figure 2.3: The Doughnut of social and planetary boundaries. Source: https://doughnuteconomics.org.

The philosophy of the doughnut approach is based on three core principles: equitable distribution of wealth, regeneration of the resources used by the economy, and creation of wealth for all people. None of these principles should have to depend on economic growth, says Raworth (2017a). In other words, we don't need to pursue unlimited growth for the economy to flourish – instead, we should pursue development that is sustainable, balanced, and equitable. The transition to the doughnut model requires a fundamental change in the way we think and act. It's about moving from a growth paradigm to sustainable development, a development that takes social justice and environmental sustainability into equal account.

2.2.1 Conclusion

Despite the Doughnut Model's focus on the economy, critics say it doesn't discuss the underlying framework conditions (i.e. the structures, rules, and institutional organization of the economy) or how money is created and managed (i.e. the financial sector). The Doughnut Model focuses on what kind of economic activity is desirable (i.e. staying within the doughnut), but it doesn't explain how to get there. These points of criticism are being addressed by the Doughnut Economics Action Lab (DEAL, n.d.), which provides tools and strategies to implement the Doughnut Model in real-world settings.

- The Doughnut model builds on the concept of planetary boundaries, adding a social dimension and including goals and degrees of goal achievement.
- In normative terms, the model prescribes that
  - social goals should be achieved without overstepping the planetary boundaries. The planetary boundaries provide the biophysical framework within which the social goals should be achieved.
  - when setting concrete goals, measures etc. at sub-global levels, the global social goals must also be taken into account.

2.3 The debate about the Great Transformation

Concepts of the "great transformation" refer to fundamental changes in social, economic, political, and ecological systems that are necessary to create a sustainable and just society.

"Great transformation" was a term used by Karl Polanyi in his 1944 analysis that the shift to a free market in the 19th century brought about profound social, economic, and political changes that fundamentally transformed people's lives and relationship with nature. Polanyi argued that unbridled market dynamics caused social and environmental problems, and that society needed to develop mechanisms to regulate and balance these problems. Similarly, today's concepts of the great transformation emphasize the importance of regulation, redistribution, and the development of alternative economic models to create a sustainable and just society. While Polanyi stressed the need for social protection measures and the importance of integrating markets into social structures, current concepts of the great transformation additionally aim to fundamentally change production and consumption patterns to reduce environmental impact.

2.3.1 Flagship WBGU report: *World in Transition*

The flagship report by the German Advisory Council on Global Change (WBGU 2011), *World in Transition,* is an important publication that addresses the challenges and opportunities of a sustainable transformation of society. It was first published in 1994 and has since been updated several times. The report analyses global environmental changes, including climate change, biodiversity loss, and resource consumption, and proposes concrete policies and social change to promote sustainable development.

World in Transition is significant to debates on the great transformation, as it argues for a far-reaching transformation of business, policy, and society. This vision extends beyond mere adaptation, demanding fundamental shifts in the structures and patterns of economic activity and daily life. The report underscores the urgency of driving forward the transition to a climate-friendly and environmentally sound economy and way of living. It emphasizes the critical role of a comprehensive sustainability policy. Finally, it identifies innovation, technology, education, institutional reforms, and international cooperation as key levers for achieving sustainable development.

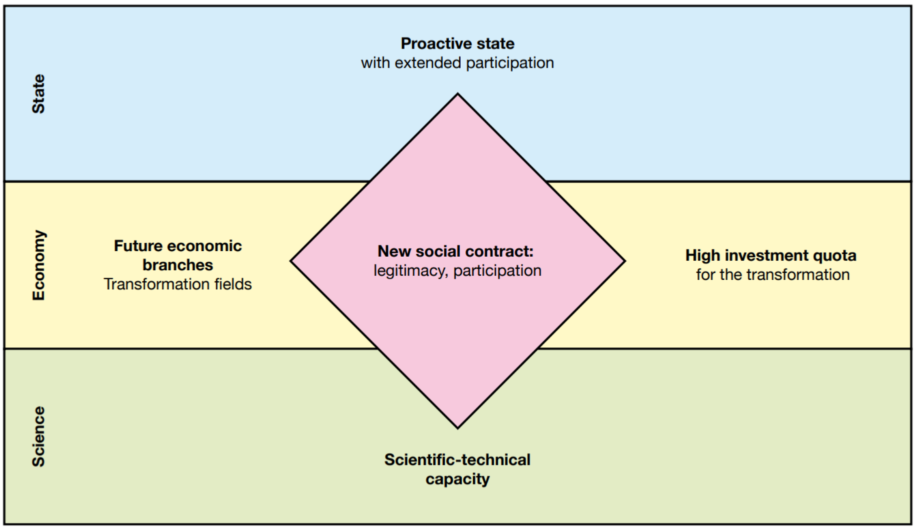

Figure 2.4: New social contract. Source: WBGU (2011, 275).

In Chapter 8, *World in Transition* distinguishes between two concepts: "transformation research and education" and "transformative research and education". These concepts are key to promoting fundamental change towards sustainability.

Transformation research analyses processes of change, particularly in the context of sustainable development. It seeks to understand the dynamics and complexity of transformations, and to identify possible courses of action for a sustainable future. Transformation research offers an inter- and transdisciplinary perspective, and it involves different actors and stakeholders in the research process.

Transformation education is education that aims to impart the knowledge, skills, and values needed to achieve a sustainable transformation of society. It includes formal and informal educational measures that enable people to actively participate in transformation processes and to promote sustainable thinking and action.

Transformative research not only generates knowledge but also goes a step further, by actively striving for change towards sustainability. It thus aims to influence social practice and develop solutions and innovations for sustainable development. Transformative research works closely with partners from practice, and strives to translate knowledge into action.

Transformative education fosters a shift in mindsets, values, and behaviour towards sustainability. It goes beyond imparting knowledge, by also promoting critical awareness, empathy, and action for sustainable changes in society.

The WBGU believes that the state should take on tasks that are not adequately performed by individuals or the private sector. It recognizes that the state plays a decisive role in creating framework conditions and shaping political measures to promote sustainable development. The state can drive innovative solutions, steer investments, establish regulations, and implement policies that enable a sustainable transformation.

> "The important thing for Government is not to do things which individuals are doing already, and to do them a little bit better or a little worse; **but to do those things which at present are not done at all**." John M. Keynes, The End of Laissez-Faire, 1926

However, the WBGU also emphasizes that effective government intervention should take place in collaboration with other actors and stakeholders, including civil society, the private sector, and academia. Jointly shaping a sustainable future is about creating partnerships and involving different expertise.

2.3.2 Schneidewind's "art of the future"

The German economist and politician, Uwe Schneidewind, discusses the concept of the great transformation in his 2018 book, *Die grosse Transformation: Eine Einführung in die Kunst des gesellschaftlichen Wandels*, which translates as *The Great Transformation: An Introduction to the Art of Social Change*. Schneidewind (2018) points out that a great transformation isn't dictated by the unstoppable development dynamics of modern societies or by a technocratic blueprint for an ecologically just society. Instead, he sees it as a process that should be actively shaped by many actors. As such, it's important to have a clearly defined normative compass and to develop the ability to navigate complex social, cultural, economic, and technological processes. He describes this ability as the "art of the future" – the skill of making desirable futures possible – and thus builds a bridge to Harald Welzer's FUTURZWEI. Welzer, a psychologist and sociologist, emphasizes the importance of imagination, creativity, and engagement in bringing about transformative change and developing alternative visions of the future.

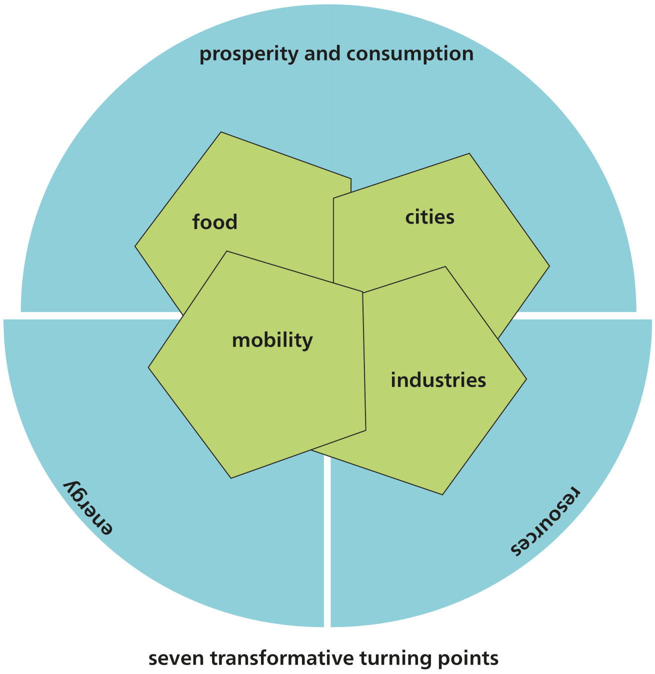

Figure 2.5: Understanding the interplay of the seven turning points of the great transformation. Source: Own illustration based on Schneidewind (2018).

For Schneidewind (2018), the great transformation has seven transformative turning points. The four dimensions of the "art of the future" (technological, economic, cultural, institutional) can be found in all seven turning points. Resource and energy transformations are fundamental. However, according to the concept of "double decoupling", they are inconceivable without a comprehensive transformation in ideas surrounding prosperity and consumption. Envisaging the desired transformation is easier in more specific sectors, such as mobility, food, cities, and key energy- and resource-intensive industries.

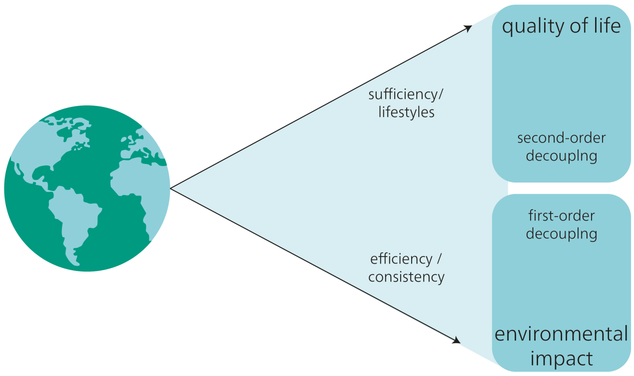

Figure 2.6: Double decoupling. Linking technology and prosperity models. Source: Own illustration based on Schneidewind (2018).

“Decoupling” refers to the separation of economic activity from environmental impact. A key concept in the sustainability debate, **double decoupling** states that sustainable development can only be achieved through increases in technological efficiency in combination with new models of prosperity and consumption. It aims to promote a more comprehensive and systemic understanding of innovation, to include both technological and social innovations.

Double decoupling refers to the following two types of decoupling:

- First-order decoupling: This focuses on increasing resource efficiency and consistency within the traditional economic growth model, mainly through technological advancements, and
- Second-order decoupling: This focuses on “sufficiency”, decoupling quality of life and a “good life” from the traditional economic growth model, as measured by GDP.

2.3.3 Global Sustainable Development Report

In September 2019, the first Global Sustainable Development Report (GSDR) was published by a group of 15 independent scientists appointed by the UN Secretary-General. Intended for publication every four years, the aim of the GSDR is to synthesize existing knowledge and identify pathways to achieving the Sustainable Development Goals (SDGs). The latest report (2023) highlights the significant gap between current progress and achieving the SDGs. Building on the 2019 report, it introduces capacity building as a new lever to accelerate progress.

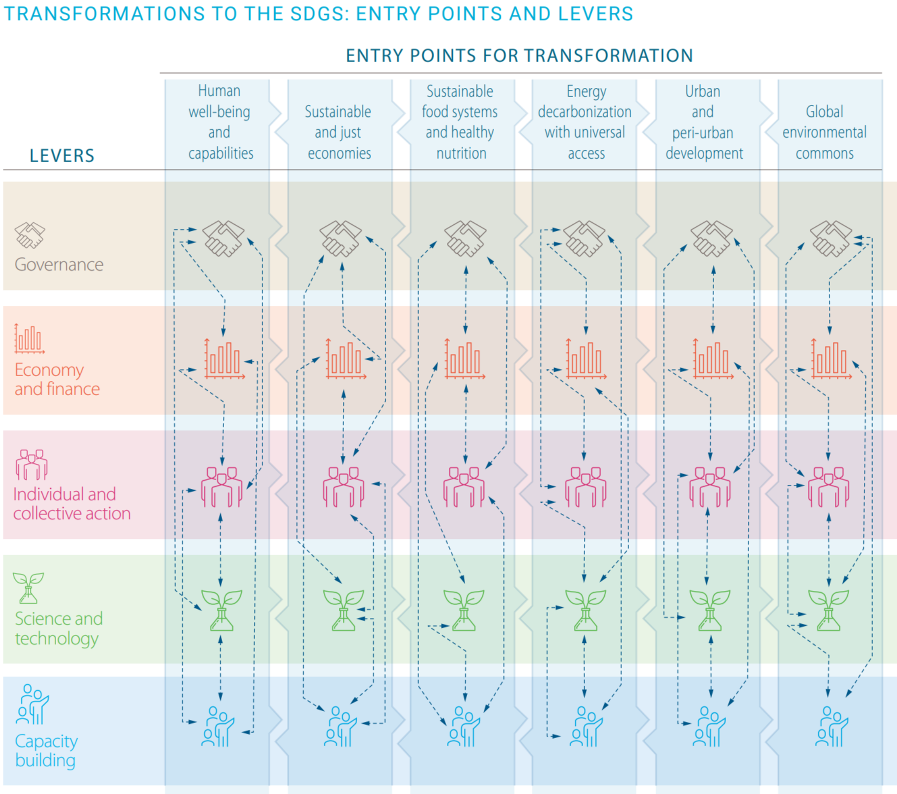

Figure 2.7: Levers and entry points for achieving the SDGs.Source: Independent Group of Scientists appointed by the Secretary-General (2023).

The GSDR proposes six key areas, or "entry points", with the highest potential to drive the large-scale and swift transformations needed. To initiate these transformations, active and multifaceted collaboration is crucial among actors from diverse fields: governance, business and finance, individual and collective action, and science and technology.

2.3.4 Conclusion

"Great transformation" concepts

- consider it necessary and possible to steer social development towards sustainability.
- are holistic concepts for managing social development. They propose entry points at various areas of society and at several levels of action.
- propose key areas with a major leverage effect (e.g. the WBGU proposes an energy transition, urban transition, land use transition, and transformative education and research).

Further reading

Polanyi, Karl. 1944. *The Great Transformation: The Political and Economic Origins of Our Time*. Beacon Press.

2.3.5 Green economy

In a pioneering use of the term “green economy”, David Pearce and Edward Barbier launched their groundbreaking *Blueprint for a Green Economy* series in 1989. The series was the first to present economic frameworks and programmatic approaches for achieving the dual goals of economic prosperity and ecological well-being (e.g. Pearce, Markandya, and Barbier (n.d.); Pearce, Barbier, and Markandya (2000); Barbier and Markandya (2013)). The green economy’s core principle challenges the traditional view that economic growth and environmental well-being are inherently at odds. It proposes a paradigm shift, moving away from bans and restrictions. Instead, it advocates for economic incentives and strategies that promote environmental sustainability while fostering positive models of economic and technological development aligned with both ecological and social goals.

In the run-up to Rio+20, the 2012 United Nations Conference on Sustainable Development, the green economy concept was finally developed into a guiding principle that shaped the debate (UNEP (2011); Bär, Jacob, and Werland (2011); Creech et al. 2012). “Green economy” was one of two key themes of Rio +20; the other was the “institutional framework” for sustainable development. Accordingly, numerous documents on the green economy were published in 2011 and 2012, and a seminal definition of the term was published by the organizing agency of Rio+20, the United Nations Environment Programme (UNEP), in the report, *Towards a Green Economy: Pathways to Sustainable Development and Poverty Eradication* (UN 2011a, p. 31; UNEP 2011, p. 16). The OECD also contributed its own response to the financial crisis with its resolution on “Green Growth” (OECD 2009a).

Note

“UNEP defines a green economy as one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities. In its simplest expression, a green economy can be thought of as one which is low carbon, resource efficient and socially inclusive. In a green economy, growth in income and employment should be driven by public and private investments that reduce carbon emissions and pollution, enhance energy and resource efficiency, and prevent the loss of biodiversity and ecosystem services.” (UNEP 2011, p. 2)

While Rio+20 cemented the concept of a green economy, the need for a fundamental global change in thinking was already widely accepted. It had become clear that while environmental protection measures cost money, they also have a positive economic impact by creating jobs (see e.g. UBA 2008) and sparking innovation in technological efficiency advancements (cf. e.g. von Weizsäcker, Lovins, and Lovins (1995)). The green economy concept systematizes such findings and calls for programmes to overcome the apparent contradictions between economy and ecology, growth and resource conservation, as well as prosperity and environmental protection. The concept thus marks a significant paradigm shift. Adhering to planetary boundaries or ecological guidelines in a green economy does not necessarily mean forgoing economic growth and technological progress ((**rockstromSafeOperatingSpace2009?**); Steffen et al. 2011; SRU 2011). Instead, the idea is that by harmonizing these apparently contradictory aims, we can achieve them even more efficiently. The EU Green Deal exemplifies this concept. The EU aims to be the first climate-neutral continent by 2050, aiming for a modern, resource-efficient, and competitive economy with net-zero greenhouse gas emissions. This plan seeks to decouple growth from resource use while ensuring a just transition that leaves no one behind.

2.3.6 Conclusion

The core thesis of the green economy is that environmental and economic goals are not contradictory. Instead, they can be reconciled through appropriate economic incentives and strategies. The green economy aims to foster both economic growth and environmental sustainability through public and private investment in low-emission, resource-efficient, and socially equitable economic systems.

Critics, however, have raised concerns about the effectiveness of green economy measures. They argue that the proposed technologies and incentives are insufficient to bring about the far-reaching systemic changes needed. An overemphasis on economic growth and technological solutions could potentially neglect essential structural and behavioural changes. And they warn that the global transition to a green economy could leave behind disadvantaged communities and developing countries, if their specific needs are not addressed.

2.4 Post-growth societies

The post-growth debate emerged from concerns raised in the 1970s, after publication of the influential Meadows Report, *The Limits to Growth* (1972). As previously mentioned, this report highlighted the Earth's finite capacity to sustain humanity in the face of unrestrained economic growth.

The post-growth debate advocates qualitative growth or even zero growth, and criticizes the effects of the "modern" economy and lifestyles (e.g. Binswanger 1985). It argues that the compulsion for constant growth is making us exceed ecological limits and leading to negative social and ecological consequences. "Ecological economics" is an important concept in this respect, as it aims to develop alternative models and approaches for evaluating economic growth.

The dilemma in this debate is that most approaches to a sustainable economy assume that a growth-independent economy should not be profit-driven. However, capitalist economies are existentially dependent on growth (e.g. M. Binswanger (2019), Oberholzer (2021)). This dilemma is key to the question of how to organize a successful transition from the current unsustainable system to a sustainable economic and social system. One main approach is to reduce dependencies on growth and promote alternative models that focus on sufficiency, without neglecting strategies that focus on efficiency and consistency. This requires changes in production and consumption patterns, in social norms, and in the political framework. The post-growth debate emphasizes the need for a comprehensive transformation that encompasses environmental, social, and economic dimensions – and aims to achieve a balance between human needs and planetary boundaries.

The concept of "sufficiency" is an integral part of the post-growth debate (Schneidewind and Zahrnt (2013)). Sufficiency aims to reduce overconsumption and promote alternative lifestyles, consumption habits, and production patterns. To promote widespread adoption of sufficiency, a legal and institutional framework that incentivizes and facilitates sufficiency-oriented practices is necessary.

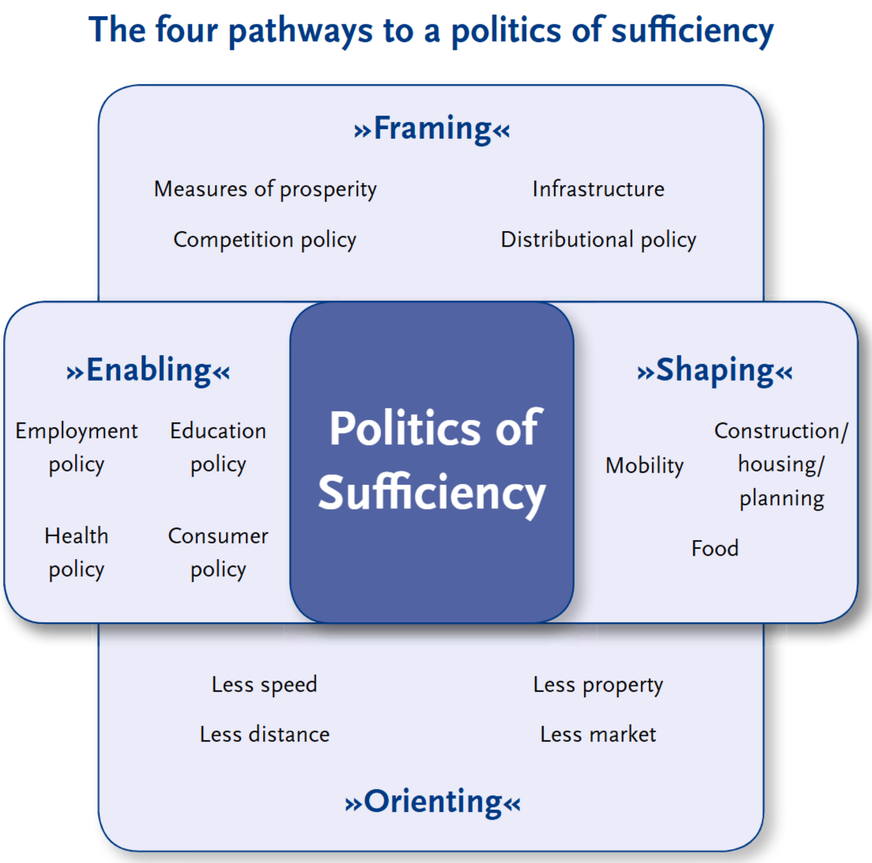

Figure 2.8: The four areas of a politics of sufficiency. (Source: Schneidewind and Zahrnt (2014))

A policy to promote sufficiency can actively shape our choices, through attractive sufficiency-oriented offers and services. It can also foster awareness and provide guidance for adopting sufficiency-oriented lifestyles and practices. This comprehensive approach aims to shift consumption towards what is necessary and meaningful, ultimately reducing excessive resource use.

2.4.1 Conclusion

Post-growth debates analyse and criticize modern society's dependency on economic growth, and the negative environmental and social effects of this growth. Rather than focus solely on technological progress and market forces, post-growth society theories strive for changes in society, structures, and institutional frameworks. Overall, post-growth debates emphasize the need for sustainable approaches to achieve a comprehensive transformation of the economy and society. Transformation to a post-growth society requires a reorientation of values, structures, and institutional frameworks. The challenge is to find ways to shape the transition to an economy independent of growth, while at the same time ensuring social justice and ecological sustainability.

Critics of post-growth debates argue that a rejection of economic growth could have a negative impact on prosperity and social progress. They fear that an economy independent of growth could lead to stagnating innovation, fewer jobs, and falling living standards. And they point out that a negative attitude towards economic growth can have potential negative effects. Constructively addressing the challenges of growth and developing viable alternatives are important aspects of enabling sustainable and equitable change.

2.5 Indicators for Sustainable Development: From Principles to Practice

A key concern of sustainable development is the ability to systematically make visible both progress and setbacks—on global, national, and local levels. Indicators play a crucial role in this effort: they help capture complex socio-ecological realities, enable comparability, and support evidence-based policymaking. Precisely because sustainability is a normative objective, particular attention must be paid to the selection, design, and use of indicators.

Importantly, indicators are not purely technical measurement tools. They inevitably reflect specific values, worldviews, and political goals. Their significance is not solely based on "objective data", but also on decisions about what should be measured, how, and for what purpose. For instance, Gross Domestic Product (GDP) has long dominated as the main benchmark for economic success—despite ignoring ecological damage, social inequality, unpaid care work, and ecosystem services.

In sustainability science and environmental policy, a useful distinction has emerged between different types of indicators, each serving a specific function (Vries, 2024):

- **State indicators** describe the condition of a system or resource—such as CO_2 concentration in the atmosphere, the proportion of protected forest area, or average life expectancy. These are primarily used to monitor trends and system states.

- **Causal indicators** capture underlying drivers or influencing factors that lead to changes within a system. These include, for example, fossil fuel consumption or socio-economic drivers like urbanization rates or consumption patterns.

- **Input or intervention indicators** refer to institutional or policy measures that are introduced in response to problems. Examples include subsidies for renewable energy, legal regulations, or public awareness campaigns. They track the presence and nature of interventions.

- **Performance or output indicators** measure the results or impacts of these interventions in relation to predefined goals. These may include emission reductions, the share of recycled materials, or progress in social justice.

This classification illustrates that individual indicators often lack significance in isolation. It is the interaction of indicators—e.g., within impact models or causal frameworks—that enables a robust assessment of sustainability trends and policy responses.

Over the past decades, a wide range of alternative indicator systems has been developed to provide a more comprehensive picture of societal development beyond GDP. These approaches aim to integrate economic, social, and environmental dimensions while making distributional and non-market contributions visible.

A significant milestone was the development of the **Human Development Index (HDI)** by the United Nations Development Programme (UNDP) in the 1990ies. The HDI combines three dimensions: life expectancy at birth (as a proxy for health), average and expected years of schooling (education), and per capita income adjusted for purchasing power (material living standard). The HDI sought to broaden the narrow monetary focus of GDP by offering a capability-based understanding of development as "the expansion of people's choices" (Sen, 1999). Today, it remains a standard reference in development economics (see UNDP Human Development Reports since 1990).

A more advanced approach is the **Genuine Progress Indicator (GPI)**, which starts with personal consumption (as GDP does) but adjusts for external costs such as environmental degradation, resource use, crime, and income inequality to reveal trade-offs between costs and benefits of economic growth. It also adds value for unpaid services like domestic work and volunteering. First proposed by Daly and Cobb (1989) as Sustainable Economic Welfare (ISEW), the GPI is well established in ecological economics (see Talberth et al., 2007). Unlike GDP, which typically increases over time, the GPI has stagnated or declined in many high-income countries since the 1980s—indicating a possible decoupling between growth and well-being.

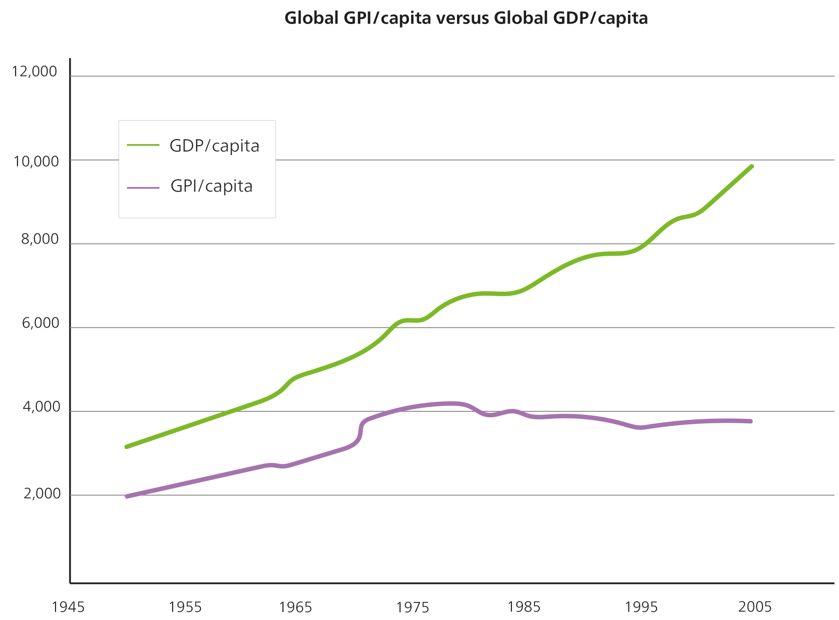

Figure 2.9: Comparison of global Gross Domestic Product (GDP) and Genuine Progress Indicator (GPI) (Source: Own ilustration)

Another commonly used measure is the **Ecological Footprint**, which translates human resource use into global hectares. It represents the biologically productive area needed to meet the resource demand of an individual, region, or country—including land for food production, settlement, and carbon absorption. Developed by Wackernagel and Rees (1996) and promoted by the Global Footprint Network, it is especially prevalent in environmental education. The Ecological Footprint illustrates whether a population is living within its ecological means—i.e., whether it exceeds its fair share of global biocapacity.

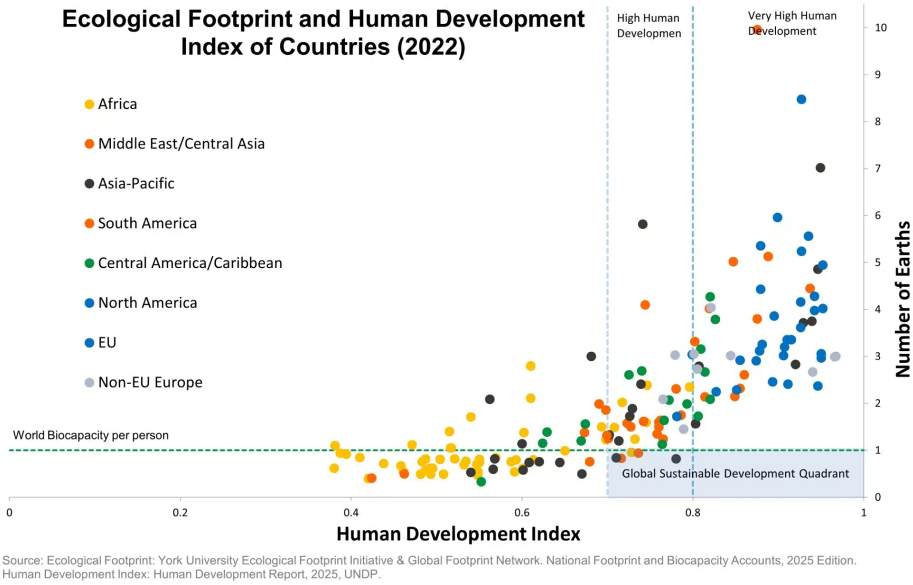

Figure 2.10: Correlation between Ecological Footprint and Human Development Index (Source: https://www.footprintnetwork.org)

The **Happy Planet Index (HPI)** offers a more subjective approach, combining life satisfaction (survey-based), life expectancy, and ecological footprint to calculate "wellbeing per unit of environmental input". Developed by the New Economics Foundation, the HPI shows that high life satisfaction does not necessarily require high resource consumption. Middle-income countries like Costa Rica often score higher than industrialized nations with unsustainable consumption patterns (NEF, 2006).

In the German-speaking world, the **National Welfare Index (NWI)** has been introduced as a GPI-based tool adapted to national data availability. It includes dimensions such as income distribution, environmental burden, and health costs. Initially developed by the Institute for Ecological Economy Research (IÖW) and the Environmental Policy Research Centre (FFU) at FU Berlin, the NWI has been tested in several German federal states (Diefenbacher & Zieschank, 2011).

Another particularly relevant measure is the **Multidimensional Poverty Index (MPI)**, developed by the Oxford Poverty and Human Development Initiative (OPHI) in collaboration with the UNDP. It captures poverty across three core dimensions: health, education, and standard of living. Indicators include child mortality, school attendance, access to electricity, and housing conditions. Unlike income-based measures, the MPI provides a nuanced understanding of deprivation across overlapping dimensions. Bader et al. (2017) demonstrate the value of the MPI in a longitudinal study on poverty in the Lao PDR, showing how rapid economic growth reduced income poverty while simultaneously exacerbating disparities in education, infrastructure, and health access.

The Ecological Footprint – Explanation and Critique

The **Ecological Footprint** measures how much biologically productive land and sea area (in global hectares) is required to sustain the resource use of an individual, population, or country over time. It includes land required for CO_2 absorption, food production, infrastructure, and other human needs. The concept was developed in the 1990ies by Mathis Wackernagel and William Rees and has become one of the most widely recognized tools for environmental communication.
The Ecological Footprint is often used in public awareness campaigns, notably in connection with the annually published *Earth Overshoot Day*—the symbolic date when humanity's resource consumption exceeds Earth's capacity to regenerate those resources in the same year.

Main limitations of the indicator based on EEA (2020):

- Non-ecological aspects of sustainability: having a footprint smaller than the biosphere is a necessary minimum condition for a sustainable society, but it is not sufficient. For instance, the ecological footprint does not consider social well-being. In addition, on the resource side, even if the ecological footprint is within biocapacity, poor management can still lead to depletion. A footprint smaller than biocapacity is merely a necessary condition for making quality improvements replicable and scalable.

- Depletion of non-renewable resources: the footprint does not track the amount of non-renewable resource stocks, such as oil, natural gas, coal or metal deposits. The footprint associated with these materials is based on the regenerative capacity used or compromised by their extraction and, in the case of fossil fuels, the area required to assimilate the wastes they generate.

- Inherently unsustainable activities: activities that are inherently unsustainable, such as the release of heavy metals, radioactive materials and persistent synthetic compounds (e.g. chlordane, polychlorinated biphenyls (PCBs), chlorofluorocarbons (CFCs), polyvinyl chloride (PVC), dioxins, etc.), do not enter directly into footprint calculations. These are activities that need to be phased out independently of their quantity (there is no biocapacity budget for using them). Where these substances cause a loss of biocapacity, however, their influence can be seen.

- Ecological degradation: the footprint does not directly measure ecological degradation, such as increased soil salinity from irrigation, which could affect future bioproductivity. However, if degradation leads to reductions in bioproductivity, then this loss is captured when measuring biocapacity in the future. Moreover, by looking at only the aggregate figure, 'under-exploitation' in one area (e.g. forests) can hide over-exploitation in another area (e.g. fisheries).

- Resilience of ecosystems: footprint accounts do not identify where and in what way the capacity of ecosystems are vulnerable or resilient. The footprint is merely an outcome measure documenting how much of the biosphere is being used compared with how productive it is.

Despite these limitations, the Ecological Footprint remains a useful entry point for engaging broader audiences in discussions on global ecological justice and planetary responsibility—especially in educational and communication settings.

2.6 Between Simplification and Complexity: Interactions Between SDGs

A central challenge in designing sustainability indicators lies in balancing simplification with complexity. Indicators are intended to support decision-making, yet must account for a wide array of interactions, uncertainties, and contextual dependencies.

This is particularly evident in the 2030 Agenda for Sustainable Development. While the Agenda includes 17 distinct goals and numerous sub-targets, these goals do not exist in isolation. Rather, they are intricately interconnected: progress in one area may reinforce, neutralize, or hinder progress in another. Consequently, indicators that illuminate these relationships are essential for coherent policy design.

Breu et al. (2020) present a methodology based on the interactions framework developed by Nilsson, Griggs, and Visbeck (2016), which systematically classifies the relationships between SDGs on a seven-point scale: from strongly negative (–3) to strongly positive (+3). They applied this approach to Switzerland's national sustainability framework, revealing areas where policy actions toward one SDG may support or conflict with others. For example, measures to reduce poverty (SDG 1) could conflict with biodiversity goals (SDG 15) if not designed to be environmentally sustainable.

Part III

Transforming (Un-)sustainable systems: Key areas and strategic approaches

Chapter 3

Environmental Sustainability

The economy is integral not only to society, but also to nature. Humans are living beings and therefore reliant on material resources to fulfill their needs. They also require intact ecosystems and a climate suitable for human habitation. When we speak of "environmental sustainability", we generally mean the protection of the diversity and functioning of natural ecosystems and the services they provide for future generations. This is an anthropocentric definition, as it focuses on human activities and how these seek to ensure the long-term sustainability of nature for human benefit. This perspective is also reflected in our everyday language and our dualistic separation of "human" and "environment", a separation that goes back to the 19th century and persists to this day. For example, we separate science into "natural science" and "social science", and our analyses often conceptualize environmental damage as "externalities". Another understanding of environmental sustainability emphasizes the importance of maintaining the self-regulation of the Earth's climate system. This view highlights the interaction and feedback between various components of the Earth's climate system.

3.1 The normative dimension of environmental sustainability

Although our understanding of how ecosystems work is based on thoroughly researched empirical information (and therefore constitutes "systems knowledge"), environmental sustainability remains a normative concept. This means that it is based not only on scientific findings, but also on an ethical evaluation (Figure **??**). The various interpretations of environmental sustainability offer different answers to the question of what should be preserved, and why. Each approach to environmental sustainability therefore expresses the desired state of the ecological environment, both now and in the future – what aspects of nature should be preserved or survive from the present to the future. For example, when we talk about ecosystem services, the intention is often to preserve ecosystems in a way that continues to support and maintain our own welfare (in this sense, it is an anthropocentric perspective).

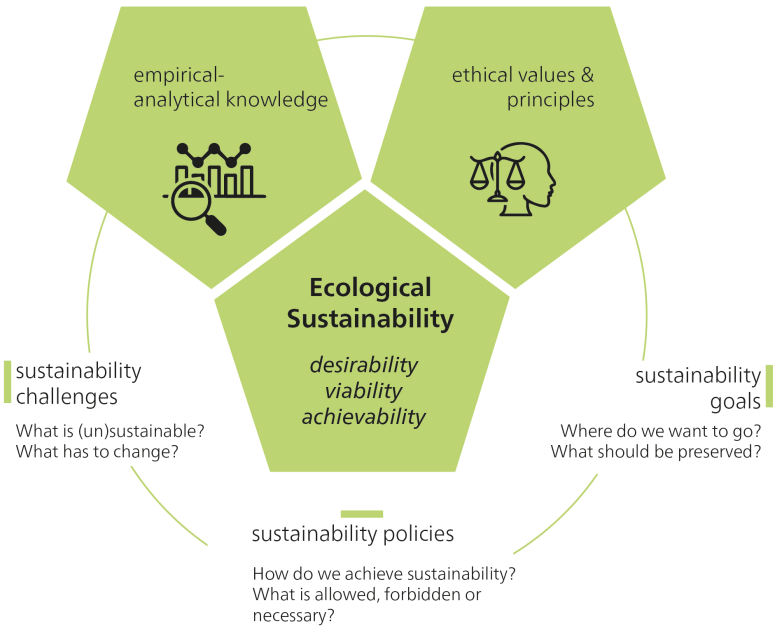

Figure 3.1: The normative dimension of environmental sustainability. Source: Own illustration.

Ethical judgments and justifications shape which ecological properties and functions we preserve for future generations, and which we deem intrinsically valuable in nature. What constitutes a desirable state for ecosystems, and why? What aspects of the ecological environment deserve protection, and what is the aim of this protection? Is it necessary to keep ecosystems as pristine as possible, and how do we define "naturalness"? To what extent should we use ecosystems for human purposes without restriction, and are there areas that we should leave for organisms to use, with minimum human intervention? Interpretations of environmental sustainability depend on the goals being pursued: for whom, why, and how (Figure **??**).

Specific examples of ethical assessments in the context of environmental sustainability

Wildlife conservation: Should we intervene in natural wildlife populations to protect endangered species and maintain the balance of ecosystems? Or should we leave nature as untouched as possible, even if this means losing some species?
Biotechnological interventions: Is it morally acceptable to use biotechnological methods, such as genetic engineering, to alter the ecological characteristics of organisms and potentially influence ecosystems?
Conservation of endangered species: Should we focus on protecting and preserv-

ing endangered species to maintain biodiversity, or should we focus on preserving broadly available species that are more important for the human diet or the economy?
Protecting ecosystems: Should we establish protected areas to preserve threatened ecosystems and species, even if this restricts local communities in their economic activities? How can we strike a balance between conservation and sustainable use?

3.2 Ecosystems, Ecosystem management, and Ecosystem services

An ecosystem is a group of living organisms and their physical surroundings that interact with each other. Ecosystems can range from small, like a flowerpot, to large, like an ocean. When similar ecosystems are found in a larger region with the same climate, they are called biomes. Energy and material flows are important for the functioning of an ecosystem. Some of these flows, such as the carbon cycle, take place at the global level, while others are more localized. Most ecosystems obtain their energy from the sun and can influence the Earth's climate through their interactions (Britannica 2025).

Ecosystem services are the benefits that humans derive from ecosystems such as mangrove forests, oceans, or wetlands. The services provided by ecosystems include the provision of clean water, the prevention of flooding, the promotion of crop growth, and the provision of places for leisure activities. The Millenium Ecosystem Assessment, published in 2005, divides ecosystem services into four categories (cf. green Box in Figure **??**): *provisioning services*, *regulating services*, *cultural services*, and *supporting services* (Millenium Ecosystem Assessment 2005). Note that the supporting services, which are primarily fundamental biophysical processes, enable and guarantee the other services in the first place.

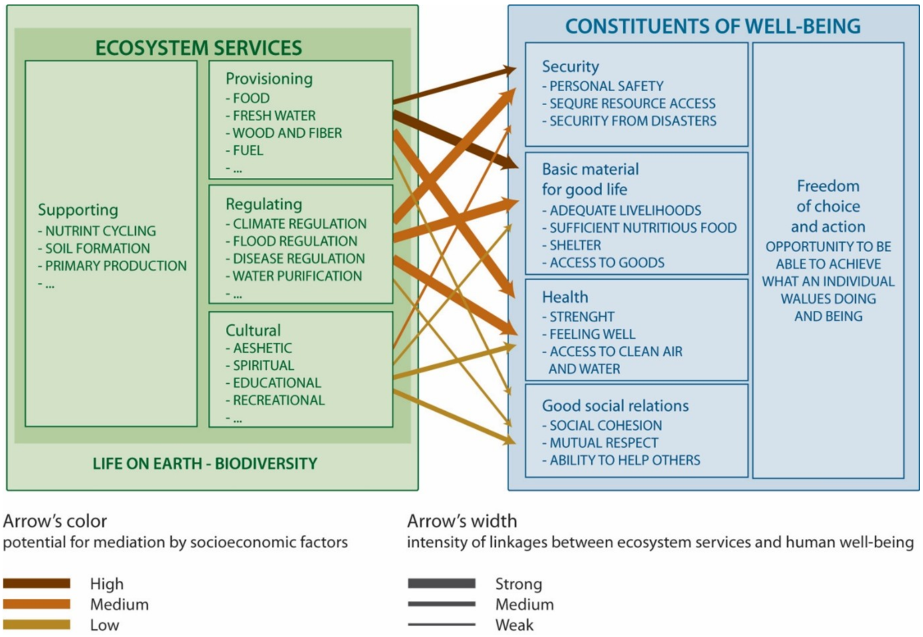

Figure 3.2: Millennium Ecosystem Assessment. Source: Millenium Ecosystem Assessment (2005). Licenced under CC.

According to the Millenium Ecosystem Assessment (2005), these ecosystem services can improve our quality of life and our economy (cf. blue Box in Figure **??**). While this way of looking at ecosystems views humans as separate from the environment, it emphasizes the relevance of ecosystem services for human well-being. Ultimately, the concept of ecosystem services seeks to place greater consideration on the value of nature in policy and economic decisions. However, the ecosystem services concept also plays a role in landscape and spatial planning. For example, it's used to analyse how spatially effective (legal) regulations impact ecosystem services, and consequently, human well-being (cf. Albert et al. 2016).

Case study: Ecosystem services in Madagascar

This landscape on the eastern coast of Madagascar offers a case study for ecosystem services. Ecosystem services in this area include firewood, which is used by local inhabitants for cooking, and rice, which is a staple food. Describing agricultural products such as these as ecosystem services illustrates that they only become "services" through human demand. Ecosystem services are not given in nature per se: instead, they represent a perspective through which humans view and describe the environment. Other ecosystem services in this landscape include the water cycle, which is vital for local inhabitants, as they rely on rivers for drinking water. Water is also used for agricultural irrigation and other purposes. Climate regulation is another significant service, important both locally and globally.
This landscape clearly illustrates that the demand for and importance of ecosystem services vary among individuals or groups. Such diverse claims on ecosys-

> tem services from the same land can lead to **conflicts of interest**. Demand for agricultural products or firewood, for example, can conflict with climate-regulating and biodiversity-promoting (international) demands on the rainforest.

Humans have had an influence on many ecosystem services, both directly and indirectly. While these impacts have often been negative, some have been positive. For example, replacing natural ecosystems with cultivated agroecosystems has increased the value of ecosystem services for humans. Take, for example, dry meadows and pastures in Switzerland. Only through centuries of forest clearing and extensive agricultural use in alpine locations could the ecosystem services these areas provided (in this case, fodder production) be used and increased. At the same time, these anthropogenically influenced sites harboured rare meadow plants and pollinators, thus fostering a high level of biodiversity. These areas are also vital for tourism, as they significantly shape the alpine landscape. Consequently, these extensively used areas generally offer greater overall ecosystem service multifunctionality than intensively farmed grasslands (Richter et al. 2024).

Many of the negative effects on ecosystems and their services result directly from human land use and overfishing. In addition, indirect factors such as climate change and invasive non-native species significantly impair the stability and functionality of ecosystems. Two research projects with significant CDE involvement – Woody Weeds and Woody Weeds+ – have demonstrated that invasive woody plants such as *Prosopis juliflora* have a major impact on biodiversity and local livelihoods in East Africa. Originally introduced to combat desertification and timber extraction, this species displaces native vegetation and is not suitable as animal feed without processing (CDE project website).

3.2.1 The monetary valuation of ecosystem management

The term *nature's services* first appeared in the publication "How much are Nature's Services Worth?" (Westman 1977). The synonymous term "*ecosystem services*" was coined a few years later (Ehrlich and Ehrlich 1981). The concept of ecosystem services and the thinking behind it are closely linked to economic philosophy and practice (Gómez-Baggethun et al. 2010). Around 20 years after the introduction of "nature's services", Robert Costanza and colleagues estimated that the **global value of 17 ecosystem services was worth 33 trillion US dollars per year** (R. Costanza et al. 1997). A recalculation in 2011 showed that this value had already fallen by between 4.3 trillion and 20.2 trillion US dollars per year (Robert Costanza et al. 2014). It is predicted that by 2050, the value of ecosystem services will either drop by up to USD 51 trillion per year, or rise by up to USD 30 trillion per year, depending on land use scenario (Kubiszewski et al. 2017).

Calculating the **monetary value of ecosystem services** typically involves three steps and requires high-quality and differentiated data. 1) The first step is to measure the change in ecosystem services after an intervention, thus placing the focus not on the status quo, but on the difference. 2) The second step is to evaluate the impact of this change on people's socio-economic well-being. 3) The third and final step is to place a monetary value on the resulting change in productivity or prosperity. There are **many different methods for calculating the monetary value of ecosystem services** (Figure **??**). One such method is “green accounting”, which uses existing market values, if available, or exchange values as an approximation (e.g. comparing avalanche barriers and protection forests). Another method is cost–benefit analysis, which can be divided into two types: stated preferences and revealed preferences.

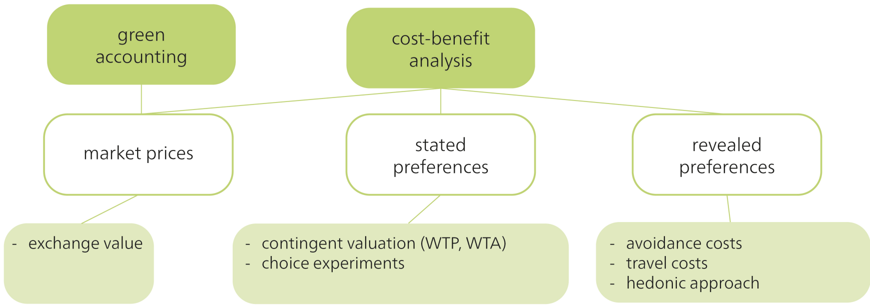

Figure 3.3: The monetary value of ecosystem services. Source: Own ilustration.

Stated preferences involves people stating (e.g. in a survey) the extent to which they would be willing to pay for certain ecosystem services, or which changes they would be willing to accept (contingent valuation). Another variation of this method involves choosing between different (usually sub-optimal) options, each representing a trade-off based on changes in ecosystem services (which are then analysed econometrically). In contrast, **revealed preferences** are based on actual behaviour. For example, it involves analysing avoidance costs (e.g. how much people pay for noise barriers) or calculating travel costs (which assumes that the time and travel costs incurred for visiting a destination reflect the value of the place, and thus, in a broader sense, an ecosystem service). Other methods include using hedonic approaches to determine the extent to which favourable or unfavourable environmental influences, and thus their quality, can affect the value of a property.

Each of these methods has its advantages and disadvantages. They are criticized mainly for their one-sided assessments and failure to account for social inequalities, power relations, and cultural differences. Consequently, these methods are limited in their ability to promote fair and environmentally sustainable decisions.

The principle behind **Payments for Ecosystem Services (PES)** involves compensating land users for maintaining a component of a specific ecosystem service. Such compensation may, for example, be paid by organizations or companies for which these specific ecosystem services are important. A well-known example of PES is the “Pagos por Servicios Ambientales” (PSA) programme in Costa Rica. This national programme, which has been operational since 1997, was set up to reduce deforestation and promote reforestation by paying landowners to maintain, reforest, or sustainably manage forest areas. The payments aim to promote the provision of ecosystem services such as carbon sequestration, water conservation, biodiversity conservation, and scenic beauty. The payments in this project come from various sources, including a tax on fuels and international funding from organizations interested in carbon offsetting. Studies show that the programme has significantly reduced the rate of deforestation in the country and promoted the conservation of biodiversity, while at the same time increasing the income of landowners (Porras et al. 2013).

3.2.2 Criticism of ecosystem services, and further and alternative concepts

While the concept of ecosystem services has been recognized and applied in the scientific and policy debate, it has also been widely criticized (see e.g. Kosoy and Corbera 2010; Norgaard 2010). One argument is that by focusing on ecosystem services, we reduce the complex and diverse values of ecosystems to their monetary benefits for humans (cf. Schröter et al. 2014). This **economic reduction of nature** neglects the non-quantifiable but nevertheless important aspects of nature, such as spiritual or intrinsic values (Fairhead, Leach, and Scoones 2012). The commodification of nature through ecosystem services could ultimately also lead to an exploitative relationship between humans and nature (cf. Schröter et al. 2014) or **an exacerbation of social inequalities between powerful actors and marginalized groups** (Dawson et al. 2021), as the concept does not explicitly consider justice (Loos et al. 2023, 478). In some cases, it is even assumed that the concept of ecosystem services could conflict with – and undermine – biodiversity conservation (cf. Schröter et al. 2014).

> "The flurry of enthusiasm for optimizing the economy by including ecosystem services has blinded us to the more important question of how we are going to make the substantial institutional changes to significantly reduce human pressure on ecosystems, especially by the rich, and to adapt to and work effectively with the rapid ecosystem changes being driven by existing and foreseeable climate dynamics." (Norgaard 2010, 1220)

In response to this criticism, the IPBES (Intergovernmental Platform on Biodiversity and Ecosystem Services), (see chapter *Biodiversity Loss*) has further developed the concept of ecosystem services and developed another concept, that of *Nature's Contribution to People* (NCP). NCPs are defined as "all the positive contributions, or benefits, *and* occasionally negative contributions, losses or detriments, that people obtain from nature." (Pascual et al. 2017, 9). The concept may resemble ecosystem services, but it goes further and recognizes that nature and its contributions to a good life can be perceived and viewed in different ways, depending on the cultural and institutional context. The concept seeks to include different world views, such as Indigenous knowledge systems, thus taking into account both intrinsic (i.e. non-anthropocentric) and relational values of nature (Díaz et al. 2018). Overall, this approach also includes context-specific perspectives, and it gives greater consideration to the justice dimensions than the concept of ecosystem services does (Loos et al. 2023).

However, such **plural valuations** are also subject to criticism and pose various challenges. For example, Jacobs et al. (2023) point out that current approaches to integrating multiple values and perspectives into ecosystem assessments risk merely promoting pseudo-participation, while existing **structures of power and discrimination** remain unchanged. Plural valuations aim for inclusivity and democracy, but they can inadvertently lead to conflict, power imbalances, and unclear outcomes if not carefully designed. Scientists have also criticized the above-mentioned NCP concept, which is often seen as an inadequate development of **utilitarian environmentalism**. This is a strongly Western and anthropocentric perspective that maintains a dualistic separation of humans and nature (cf. Muradian and Gómez-Baggethun 2021). Muradian and Gómez-Baggethun (2021) [p. 7], propose that "in order to induce transformative change in human-nature relations we need a shift from a morality of utility to a morality of care, a reallocation of property rights, and the extension of the community of justice to non-human entities." A notable example of this transformation is the 2017 recognition of New Zealand's Whanganui River as a legal person (s. Charpleix 2018).

3.3 Planetary boundaries

The current era is called the Anthropocene because human activities are exerting an increasingly profound impact on the Earth and its systems. Although this term has not (yet) been recognized as an official geologic or geochronological epoch by geological bodies (Quelle: Zhong 2024), it is used to describe the enormous human impact on the environment. There is no uniform definition of the origins of the Anthropocene; possible markers include human impact on greenhouse gas concentrations through rice cultivation, the global spread of radioactivity from nuclear testing, or the appearance of plastic particles in geological sediments. The Industrial Revolution is generally considered the start of the Anthropocene.

A quantitative definition of environmental sustainability was proposed by Rockström et al. (2009) and Steffen et al. (2015), who identified nine planetary boundaries: climate change, biosphere integrity, land-system change, freshwater change, biogeochemical flows, ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, and novel entities. These planetary boundaries serve as a scientific reference point and show the limits within which our global socio-economic systems must operate to ensure the well-being of humanity (see also the Debates section).

This chapter focuses on three of these boundaries, described here as **climate change**, **land use change**, and **biodiversity loss**. These three crises are mutually reinforcing and form a complex web of interactions.

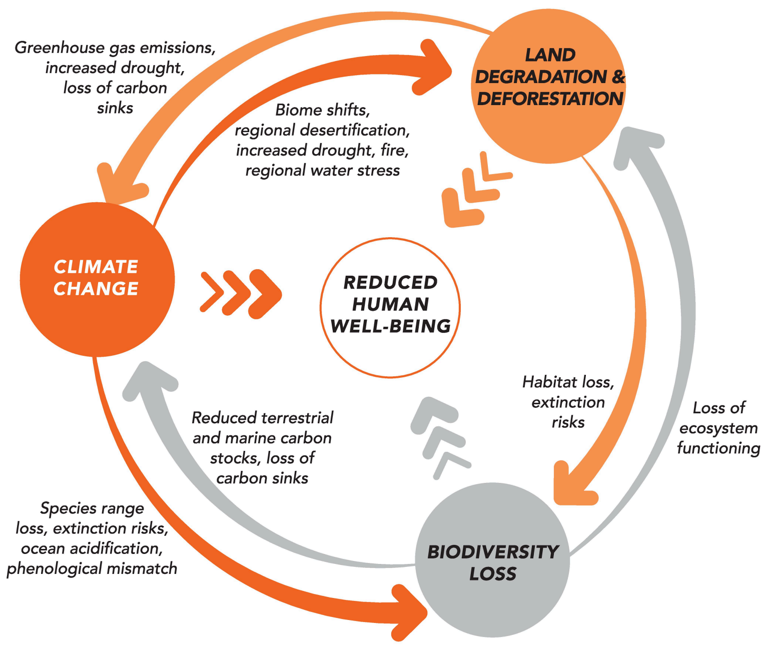

Figure 3.4: Interactions between biodiversity, climate change and land use. Source: UNEP (2021b), CC

3.3.1 The climate crisis (climate change)

Climate change – or, to use a term more illustrative of the urgency, the climate crisis – is one of the most pressing issues of our time. It receives much media attention and is exemplary of a “wicked” problem (see page 13; Hulme, 2009). It is the focus of the Fridays for Future movement that was initiated by Greta Thunberg, and it is a topic, of discussion at least, on the (global) political stage. While local conditions and natural fluctuations contribute to some uncertainties in climate models and statistics, the scientific evidence clearly shows that rising global temperatures have an enormous impact on environmental sustainability and human well-being. This is why climate change is one of the nine planetary boundaries.

The term **climate** describes the average state of the atmosphere (temperature, precipitation, humidity, etc.) over a period of at least 30 years at a specific location. By contrast, **weather** describes the short-term state (i.e. from seconds to weeks) and includes weather phenomena such as thunderstorms or heat waves (source - NCCS.admin). The **climate system**, in turn, encompasses not only the atmosphere but also the hydrosphere (oceans, lakes, rivers, groundwater), the cryosphere (glaciers, snow cover, sea and lake ice, permafrost), the biosphere (living organisms), and the lithosphere (soils or the outermost layer of the earth). In particular, it refers to the physical, biological, and chemical interactions and interdependencies between these five spheres. In addition, the climate system is influenced by radiation from space (see figure below).

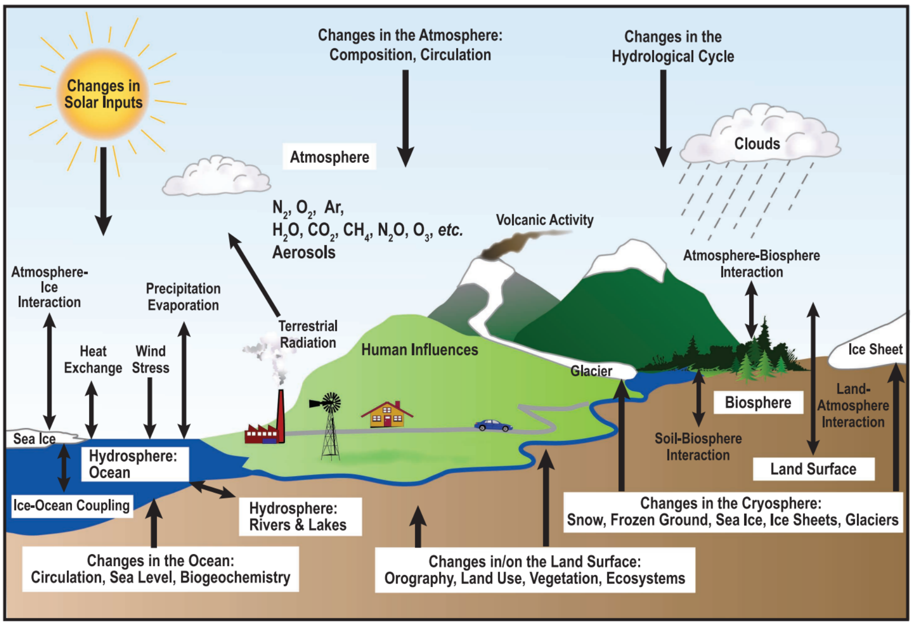

Figure 3.5: The components of the climate system, their processes, and interactions. This includes the atmosphere, biosphere (life), cryosphere (ice), hydrosphere (water), land and human influences, and the interactions between these components (shown with two-way arrows). The parts of the system that can change are labeled as "Changes". Source: IPCC AR4 FAQ

Material and energy flows between the various components of the climate system, such as the carbon cycle, can act as **feedback mechanisms**. We have already learnt about such feedback mechanisms in the chapter on systemic effects (see *chapter 2*). These can either be positive, if the feedback reinforces the original change, or negative, if the feedback attempts to reverse the original change. Here are examples of a positive and negative feedback effect in the climate system:

- **Positive feedback**: When the Earth's surface temperature increases, so does water evaporation, creating a positive feedback loop. Water vapour is the most significant **greenhouse gas** in the atmosphere. As temperatures rise, more water vapour enters the atmosphere, increasing the reflection of heat back to the Earth's surface and leading to further evaporation. This cycle accelerates warming and atmospheric water vapour accumulation. On Venus, this feedback mechanism intensified the greenhouse effect to the point where all liquid surface water evaporated.

- **Negative feedback**: Increased atmospheric CO_2 can enhance photosynthesis, a process known as **CO_2 fertilization** and an example of a negative feedback mechanism. Although an increase in carbon dioxide can cause plants to bind more carbon in biomass than before, it is far from sufficient to compensate for the increasing CO_2 content in the atmosphere, which is mainly caused by the burning of fossil fuels and changes in land use. The ability of plants to take up additional CO_2 is limited by various factors such as nutrient availability, water resources, and temperature conditions. Therefore, this feedback mechanism is insufficient in compensating for the increase in atmospheric CO_2 concentrations caused by human activities.

Positive feedback mechanisms or the failure of negative feedback can lead to significant *thresholds* being exceeded in the climate system. These thresholds are referred to as **tipping points of the climate system** and are closely linked to the concept of planetary boundaries (see *chapter 2.4 on The debate about the Great Transformation)*. When these thresholds are exceeded, abrupt and often irreversible changes can occur in the climate system, destabilizing the climate and leading to accelerated climate change. The concept of tipping points emphasizes the urgency of limiting global warming and reducing greenhouse gas emissions. Exceeding these thresholds will make it increasingly difficult to control climate change or minimize its effects. Ultimately, the Earth risks becoming a large greenhouse (*Hothouse Earth* scenario) (see Figure **??**).

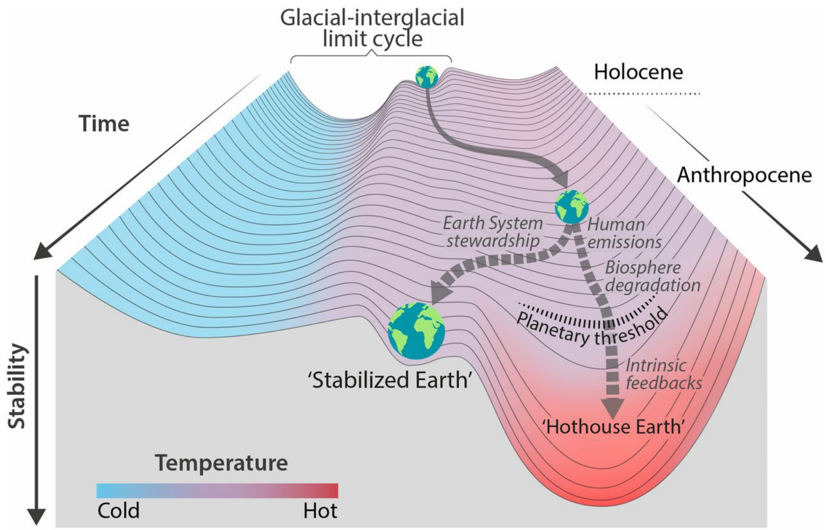

Figure 3.6: Stability landscape showing the pathway of the Earth System out of the Holocene and thus, out of the glacial–interglacial limit cycle to its present position in the hotter Anthropocene. Source: Steffen et al. (2018), CC BY-NC-ND 4.0

A total of 16 tipping points are assumed to exist (Figure **??**), including the Siberian permafrost soils or the West Antarctic Ice Sheet. These tipping points have different threshold values, meaning that some will be reached earlier, with less global warming than others. If these tipping points are reached or exceeded, they can trigger self-reinforcing feedback effects that lead to further warming, initiate further tipping points, and intensify climate change. Some of these tipping points are already in a critical state, as explained impressively by Johan Rockström, one of the founders of the Planetary Boundaries concept (https://m.youtube.com/watch?v=Vl6VhCAeEfQ). In the Amazon, for example, some areas now emit more CO_2 than they absorb – becoming carbon *sources* rather than sinks – due to forest fires, climate-change-induced water stress, and deforestation (see Land use chapter) (Gatti et al. 2021; Harris et al. 2021). Consequently, the Amazon is already in transition and losing resilience (Flores et al. 2024). Another worrying and very real tipping point is the *Atlantic Meridional Overturning Circulation*, or AMOC (Rahmstorf, 2024; https://www.youtube.com/watch?v=ZHNNW8c_FaA). The AMOC is nearing its tipping point (van Westen et al. 2024), with projections suggesting it could collapse as early as the middle of this century (Ditlevsen & Ditlevsen, 2023). A collapse would lead to increased European heat waves and sea-level rise along the north-eastern US coast (Rahmstorf, 2024).

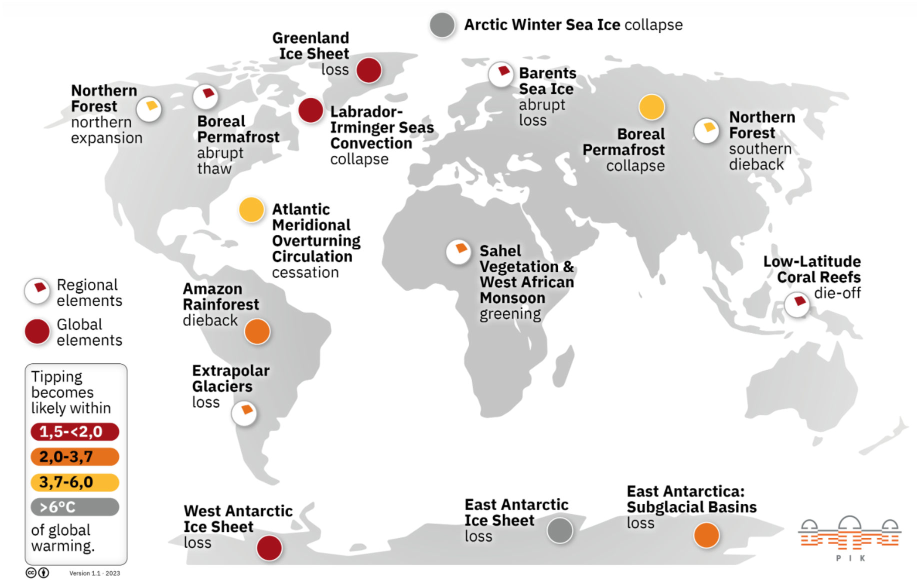

Figure 3.7: The 16 tipping points and their likelihood of occurring due to global warming. Source: https://www.pik-potsdam.de/en/output/infodesk/tipping-elements/tipping-elements, based on Armstrong McKay et al. (2022).

But what is causing the Earth to warm and climate change to occur? In addition to natural causes of climate change such as geological, long-term causes (Earth's axial rotation, Milanković cycles, volcanic eruptions), the primary driver of global warming is the **greenhouse gas effect**. The greenhouse gas effect is a process where short-wave solar radiation enters the Earth's atmosphere and is re-emitted from the Earth's surface as long-wave infrared radiation. Greenhouse gases – such as carbon dioxide, methane, and water vapour – absorb some of this infrared radiation and emit it in all directions, causing some of the heat to return to the Earth's surface. Without this effect, the Earth would be too cold to support life. Today, however, there are too many greenhouse gases in the atmosphere due to anthropogenic emissions, causing the earth to warm up.

The largest share of human CO_2 emissions, around ¾ of the global total, originates from energy production, i.e. primarily electricity and heat generation (Figure **??**). This is largely due to the combustion of coal, gas, and oil, with emissions from industry (comprising 24.2% of the global total), transport (16.2%), buildings (17.5%), and agriculture (18.4%). The main drivers of agricultural emissions are livestock farming, deforestation for agricultural land, and the use of nitrogen fertilizers on agricultural soils.

Within the transport sector, air traffic accounts for 1.9% of global emissions (see figure below). In Switzerland, however, flights account for about 11% of national emissions. These flight emissions are largely attributable to international air traffic (FOEN, 2025), although the calculations exclude Basel-Mulhouse Airport, which is located on French soil. Despite this evidence, Switzerland has not yet introduced a CO_2 tax on kerosene (the main component of aviation fuel), although there is one on fossil fuels such as heating oil and natural gas. It should however be noted that climate impact is not solely determined by greenhouse gas emissions (Allen et al., 2018; Lee et al., 2021; Neu, 2021). The warming potential of air travel can thus be significantly higher than indicated by greenhouse gas emissions alone, as non-CO_2 effects from flights – such as water vapour, nitrogen oxides (NOx), and soot – also affect the climate (Lee et al., 2021).

Globally, air travel emissions have more than doubled since 1990, driven primarily by international flights. Despite a sharp drop in the number of flights in 2020 due to the pandemic, flight emissions have rebounded and are projected to surpass 2019's record levels in the coming years.

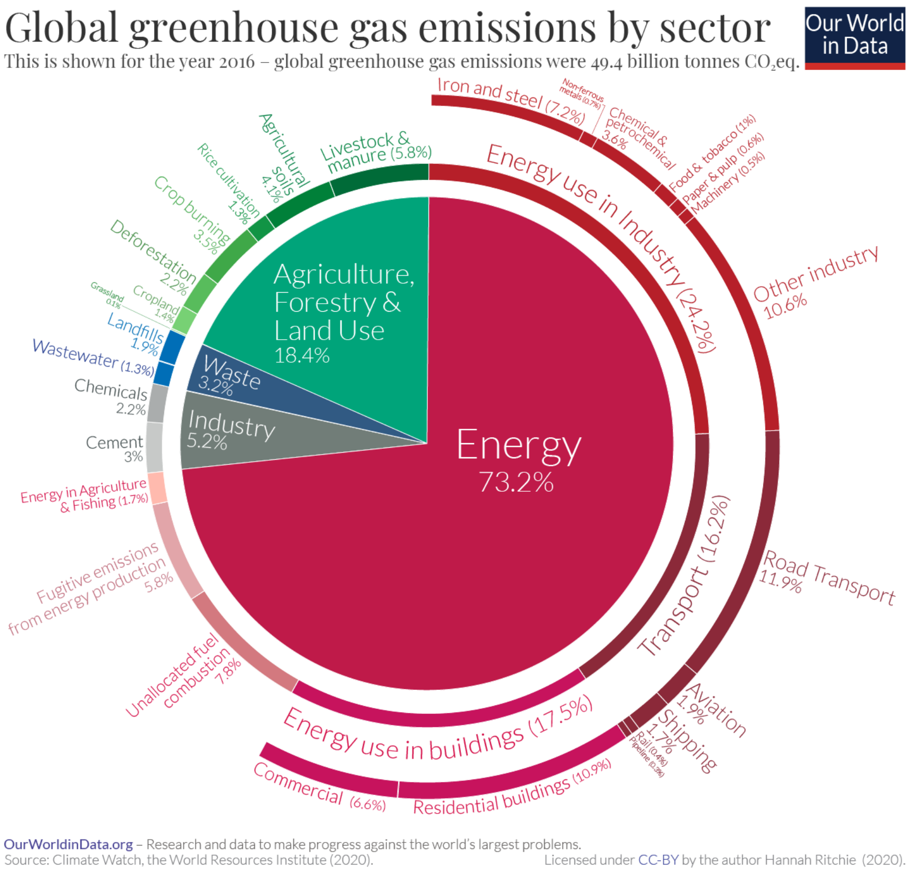

Figure 3.8: Global greenhouse gas emissions by sector (Source: Climate Watch, the World Resources Institute (2020))

Recent calculations and climate reports indicate that the Earth's temperature has risen by around 1.2°C in the last 10 years (2014–2023) compared to pre-industrial levels (1850–1900). In addition, 2023 was the warmest year on record, with a global surface temperature of 1.45°C above the pre-industrial baseline. This may not sound like much, but it represents a substantial heat increase, given the oceans' vast size and heat storage capacity. (https://www.un.org/sites/un2.un.org/files/temperaturerise_may2024.pdf) (https://ourworldindata.org/ghg-emissions-by-sector)

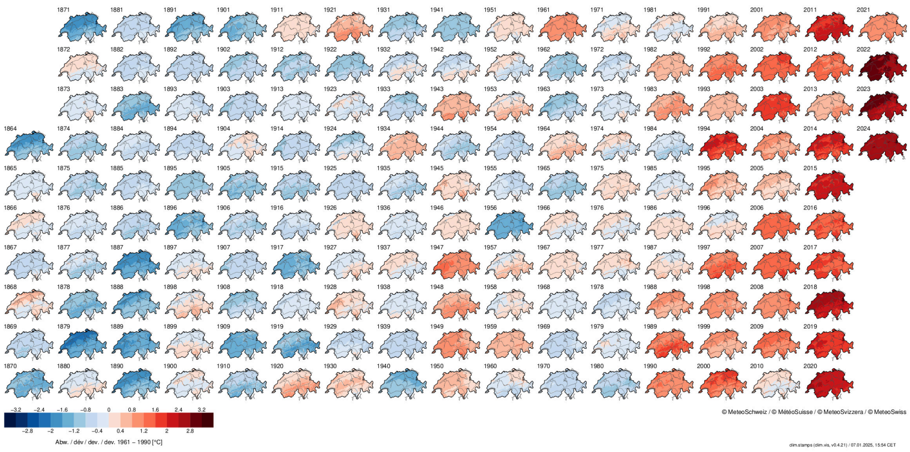

Figure 3.9: What is climate (Source: https://www.nccs.admin.ch/nccs/en/home/climate-change-and-impacts/climate-basics/what-is-climate-.html)

The **Paris Climate Agreement**, adopted in December 2015, was the first international treaty where industrialized and emerging countries jointly pledged to curb greenhouse gas emissions. Its core objective is to limit global temperature rise to a maximum of 2°C, with efforts to pursue a 1.5°C limit, relative to pre-industrial times. In 2018, the Intergovernmental Panel on Climate Change (IPCC, see box) published a Special Report containing some startling facts: Most notably, that we only have few years left – until 2030 – to avoid exceeding 2°C. The next major decision in international efforts that built on the Paris Climate Agreement was the 2022 climate conference, COP 27. At COP 27, signatories agreed to establish the Loss and Damage Fund, a result of decades of prior discussion, designed to help developing countries to receive financial support after extreme weather events.

The Intergovernmental Panel on Climate Change (IPCC)

The **Intergovernmental Panel on Climate Change (IPCC)**, established in 1988 by the United Nations and the World Meteorological Organization, synthesizes scientific research on climate change for policymakers worldwide. Its periodic IPCC reports provide a comprehensive assessment of the current state of knowledge on climate change, including the physical foundations, impacts, risks, and adaptation strategies. These reports are based on thousands of studies and are recognized as the most authoritative source of information on climate change. The IPCC reports serve as the basis for international negotiations and national climate policies, including the Paris Agreement of 2015.